

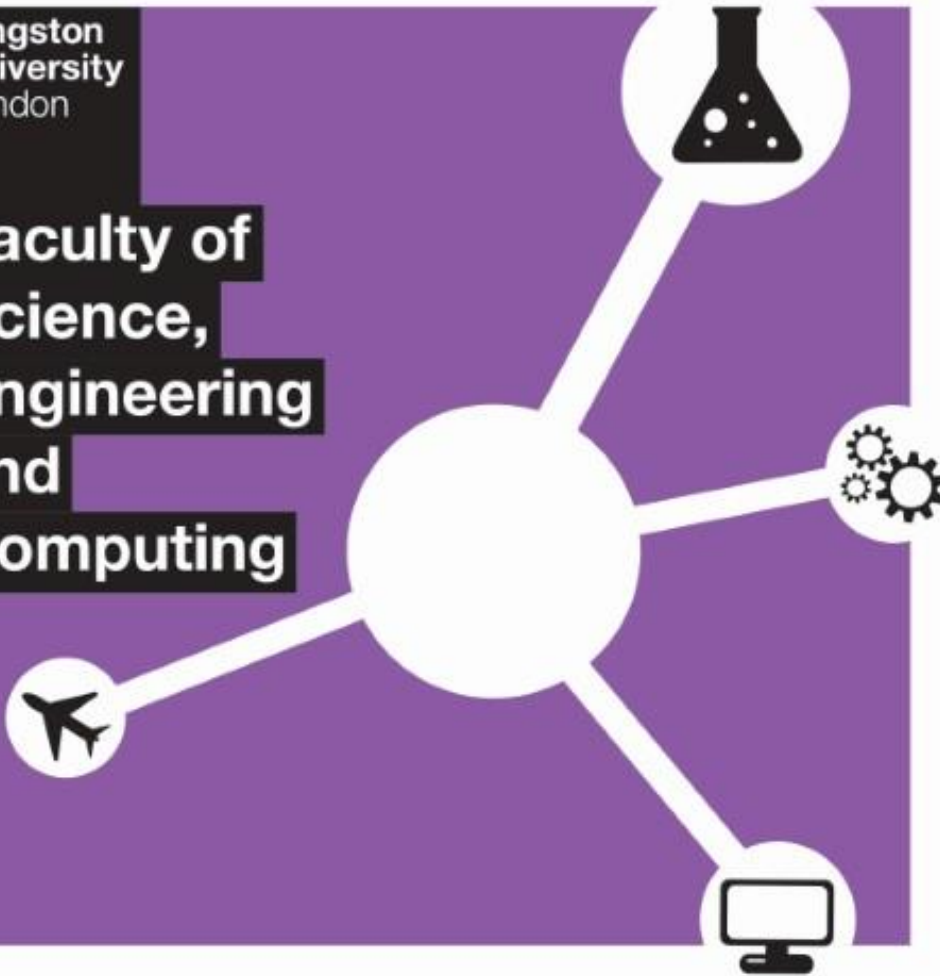
Class Props

- **None**

House Keeping

Kingston
University
London

Faculty of
Science,
Engineering
and
Computing



KEEP
CALM
AND
TURN PHONE
TO SILENT MODE

Attitude Orbit Control Systems



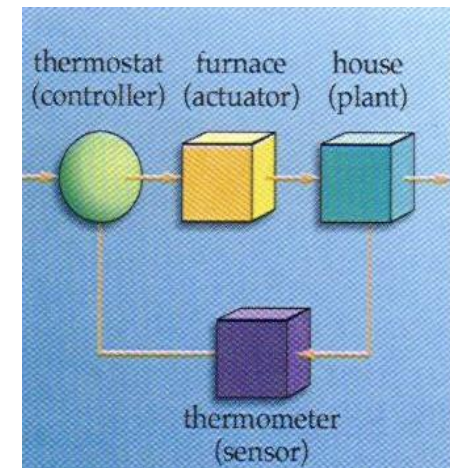
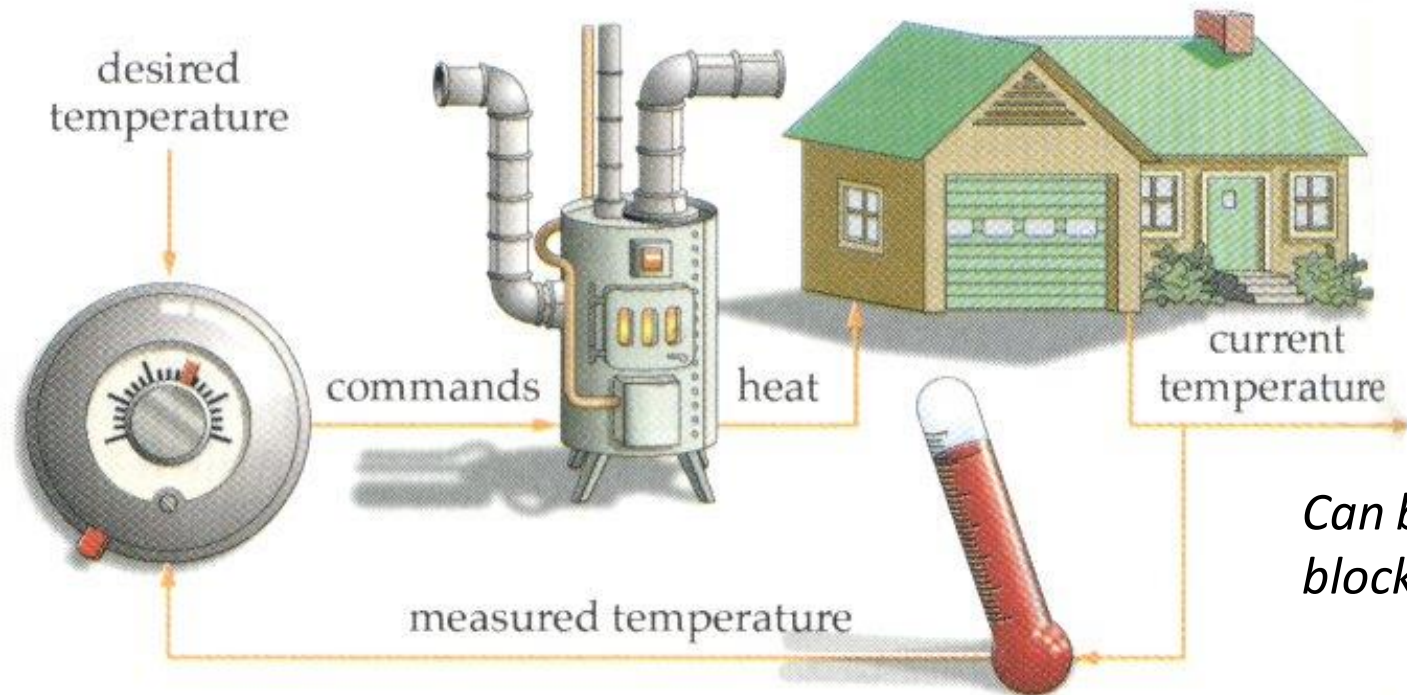
AE6030 Space Vehicle Design



Attitude Orbit Control Systems

Control Engineering / Systems

The purpose of a control system is to design a controller for a system to achieve a desired behaviour.



*Can be represented by
block diagrams*

Open and Closed Loop Control

Open loop

- Compute control input without continuous variable measurement
 - Simple
 - Need to know EVERYTHING ACCURATELY to work right
- Open-loop control fails when
 - We don't know everything
 - We make errors in estimation/modelling
 - World / things change

Attitude Orbit Control Systems

Open and Closed Loop Control

Closed loop

- Measure variables and use it to compute control input
 - More complicated
 - Continuously measure & correct
 - Cruise-control car: measure speed & change engine force
 - Sounding rocket: Measure orientation and rate relative to vertical and position canard fins to change attitude
- Feedback control: control possible, *even if*
 - We don't know everything
 - We make errors in estimation
 - Things change



Attitude Orbit Control Systems

Closed Loop Control

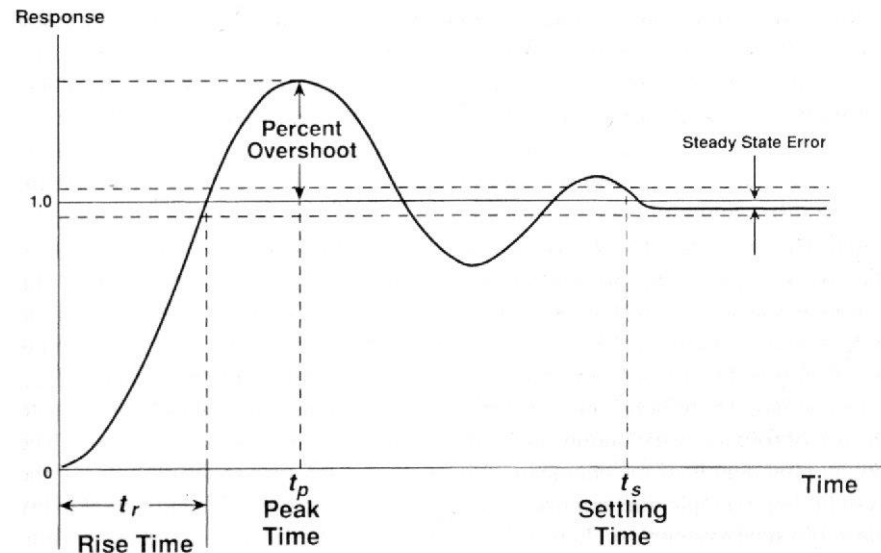
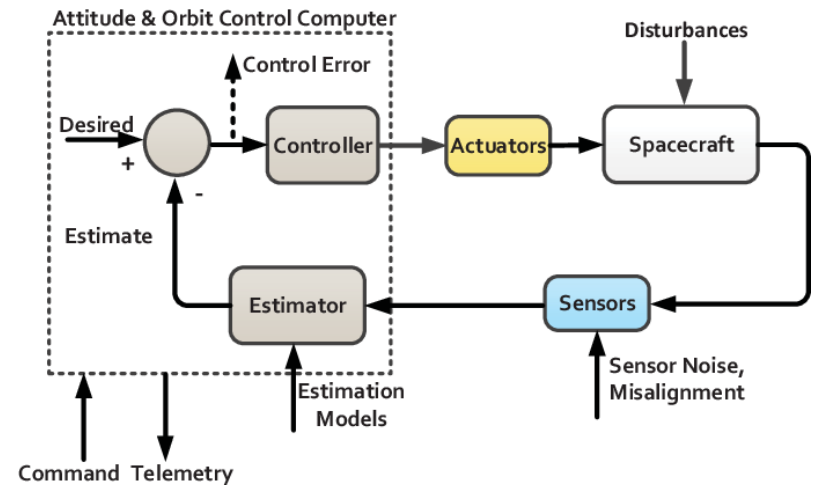
All control systems must:

Understand the system behaviour (react to inputs to produce outputs through a *model*) i.e. Know how to stabilise the spacecraft

Observe - sense the system behaviour i.e. measure the spacecrafts attitude

Decide what to do i.e. Reconstitute the attitude from measurements and determine a course of action

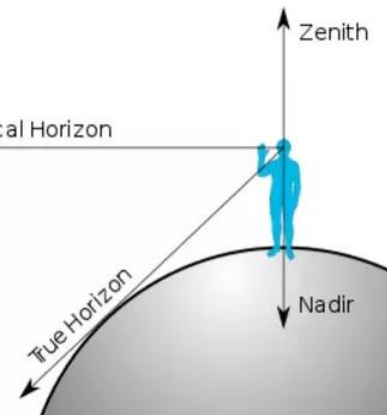
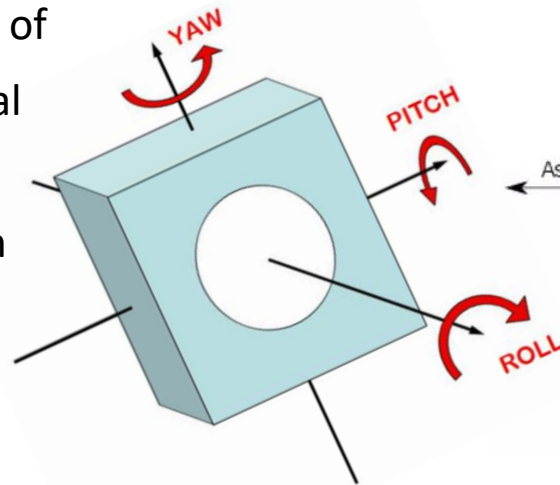
Effect a response – actuators i.e. Re-orientate the spacecraft



Attitude Orbit Control Systems

Quick Interlude: Terminology and Definitions

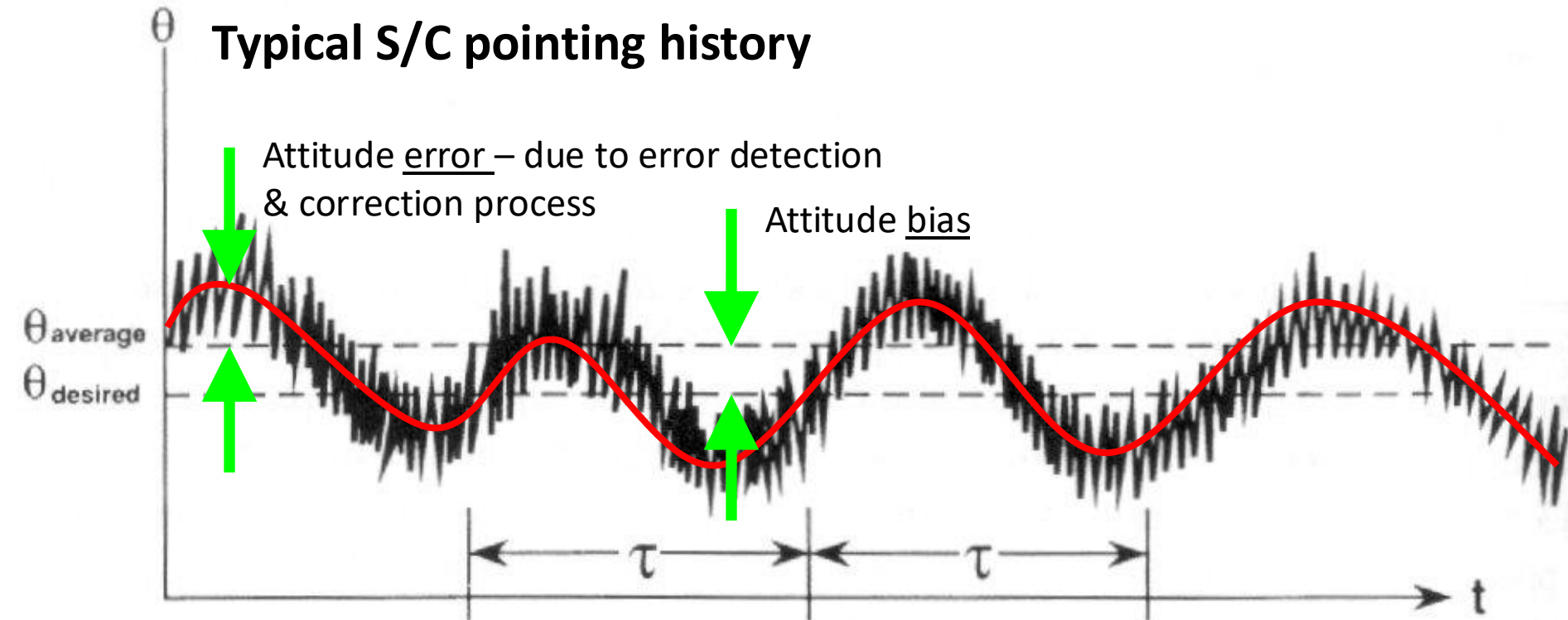
- **Attitude** (not altitude) – The orientation of a defined spacecraft body coordinate system with respect to a defined external frame.
- **Attitude control** - Maintenance of a desired, specified attitude within a given tolerance
- **Attitude determination** - Real-time calculations, within a given tolerance, of the spacecraft attitude
- **Attitude error** - “Low Frequency” spacecraft misalignment; usually the intended topic of attitude control
- **Attitude jitter** - “High Frequency” spacecraft misalignment; usually ignored by the Attitude Determination and Control System (ADCS); reduced by good design, internal damping or fine pointing/optical control.
- Nadir and Zenith



Attitude Orbit Control Systems

Quick Interlude: Terminology and Definitions

Typical S/C pointing history



When desired attitude = constant $\rightarrow$ pointing

When not constant $\rightarrow$ tracking, slewing, rotating

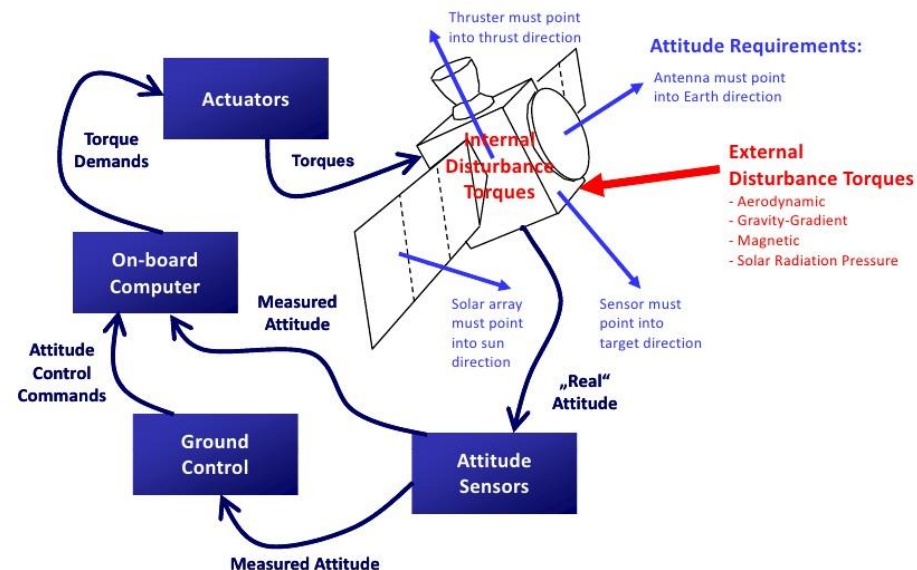
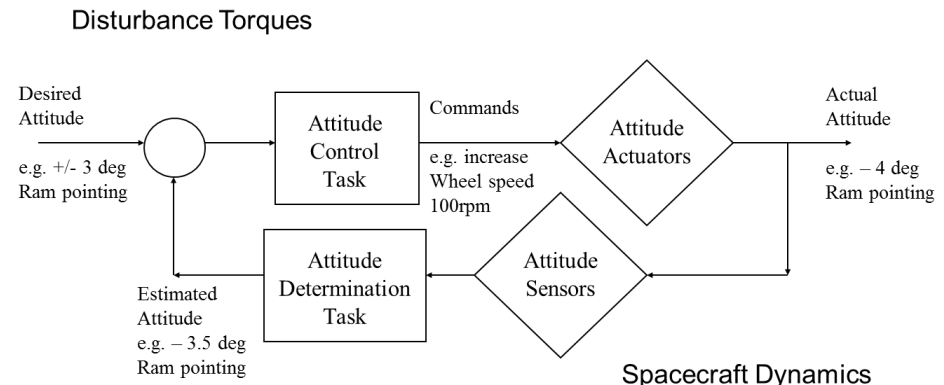
Attitude Orbit Control Systems

Basic Concept of Attitude Control

1. **Measure the satellite attitude** using attitude sensors (i.e. GPS, Magnetometers, Sun Sensors, Earth sensors, star trackers etc.) and compare to the desired attitude. The difference is known as the attitude error.
2. Use the attitude error to **estimate the required corrective torque** needed to be generated by the spacecraft.
3. **Apply the corrective torque** using the spacecraft actuators (i.e. magnetorquer rods, thrusters, reaction wheels, booms etc.).
4. Measure the new attitude and **repeat the cycle** ...

Why repeat continuously?

1. The spacecraft is continuously **under ever changing disturbance** torques (i.e. Earth shape, Sun/Moon gravity, Solar Radiation pressure etc.).
2. **Calculations are estimates**, they are not absolutes so the corrective torque will continuously over or under shoot the required torque.
3. Actuators have their own limitations and **may not provide the exact torque required** i.e. Thrusters have a delay when opening and closing valves leaving a finite time (and hence impulse) that will always be delivered, magnetorquer rods provide magnetic impulses that are hard to fine tune.



Attitude Orbit Control Systems

Case Study: Launcher Deployment



Iridium-8 satellites deployment

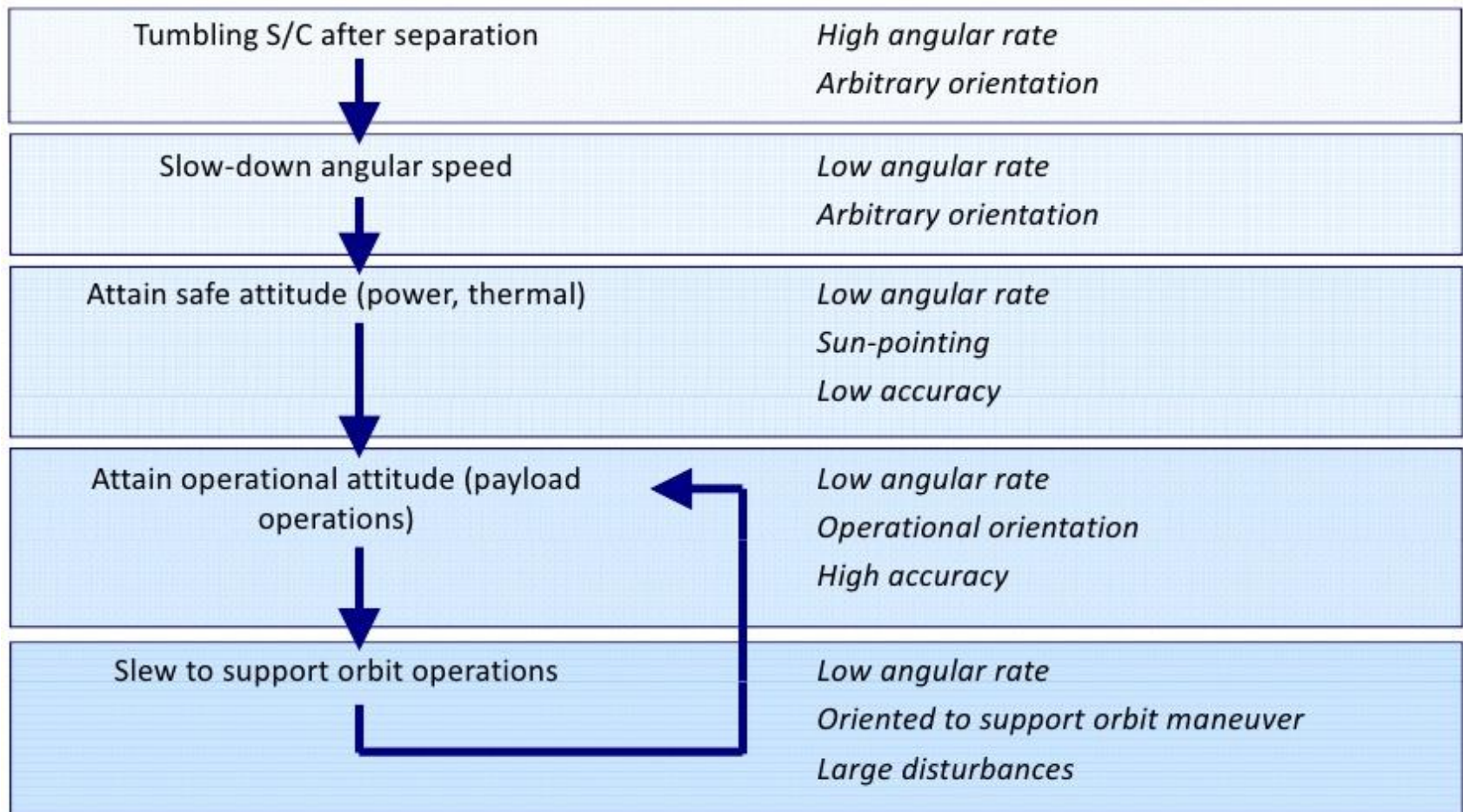


PSLV-C37 deploys 88 Planet CubeSats

Note that due to crude spring loaded deployment mechanism, tumble rates are much higher!

Attitude Orbit Control Systems

Case Study: Launcher Deployment

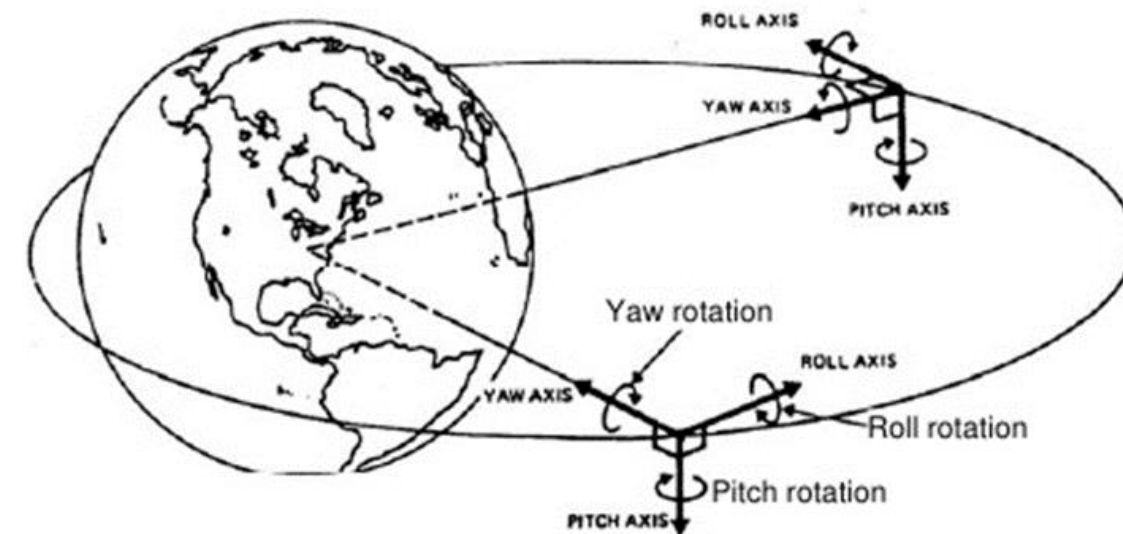


Attitude Orbit Control Systems

Description of Spacecraft Attitude

To give a description of the attitude and describe a spacecrafts orientation a fixed coordinate system (spacecraft centric) should be defined with respect to a reference coordinate system (global centric).

It is **vitaly important** that all subsystem engineers know what these co-ordinate systems are. A key responsibility of a System engineer is to control and manage this. An error here can cause mission failure. (i.e. detumbling of S/C with wrong co-ordinates could increase the tumble rates)



An Example

The global centric coordinate frame could be aligned to the orbit. I.e. Orbit plane.

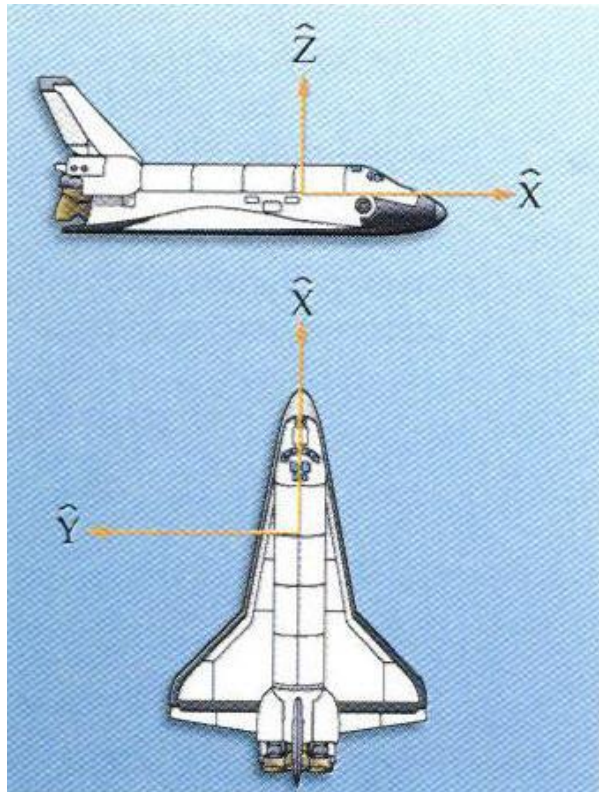
The spacecraft centric coordinate frame could be aligned to the pitch, roll and yaw axis of the spacecraft.

The two could be defined with respect too each other by;

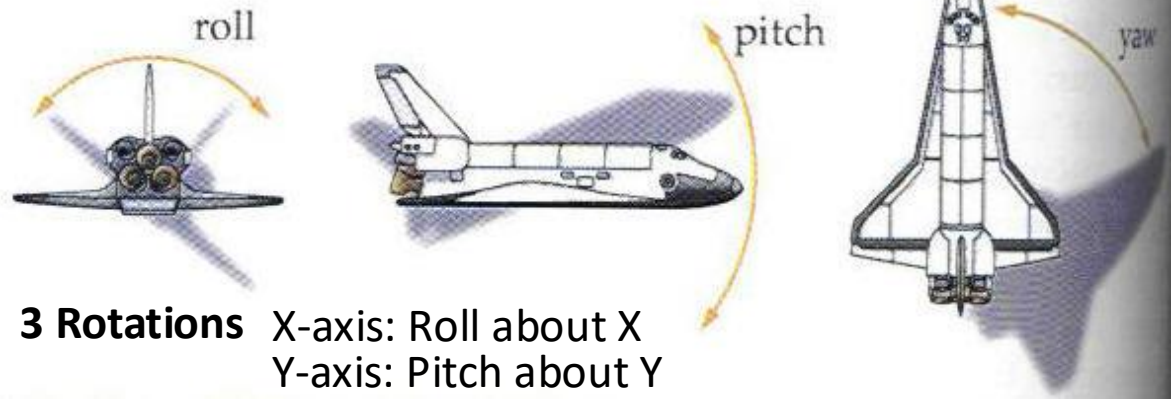
- **Yaw axis** – points to Nadir (Earth pointing)
- **Pitch axis** – points to South orbit normal
- **Roll axis** – perpendicular to yaw and pitch

Attitude Orbit Control Systems

Description of Spacecraft Attitude: Further Examples



Body Frame
X-axis: nose
Y-axis: left wing
Z-axis: top



3 Rotations
X-axis: Roll about X
Y-axis: Pitch about Y
Z-axis: Yaw about Z

CoM (Centre of Mass)

Box-shaped Spacecraft don't have noses or wings. So, the CoM is defined as the coordinate origin (as this will be the point where the spacecraft will rotate about). However, the axis will be defined separately and could be based on;

- Solar panel location
- Communication antenna location
- Payload location
- Etc.

Attitude Control Requirements: Pointing Accuracy

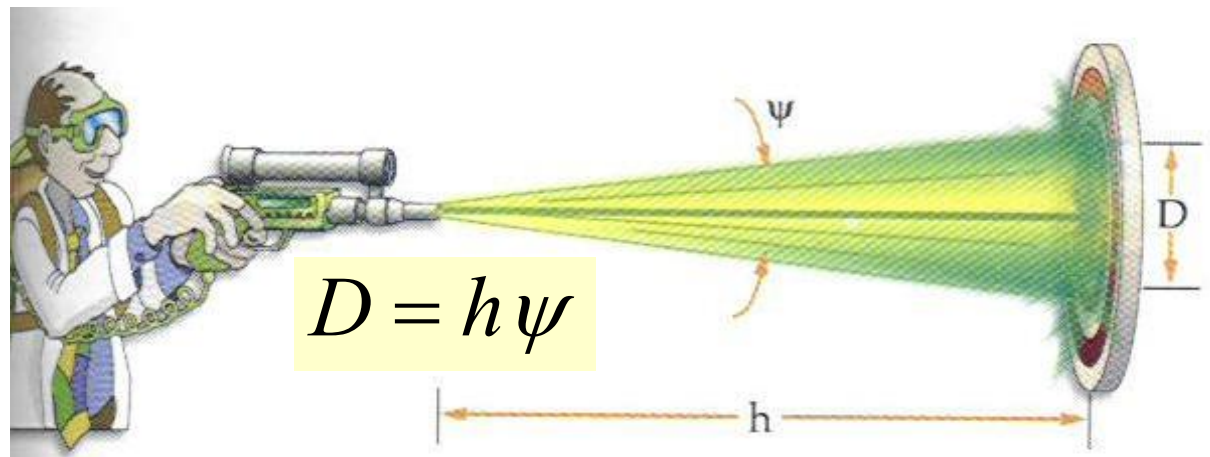
Now that we know how to describe attitude, how do we know if a S/C has the right attitude?

Attitude control is required when we want to control the attitude within a certain tolerance (defined by mission/system requirements). To know if we are operating within that tolerance we need know how accurate our system is.

Laser beam example

Attitude or pointing accuracy depends on the size of the target and how steady the laser is held. The smaller the target, the steadier the laser must be held.

The laser beam will tend to stay within a cone, centered on the target, this cone can be given an angular size, ψ , which defines the pointing accuracy.

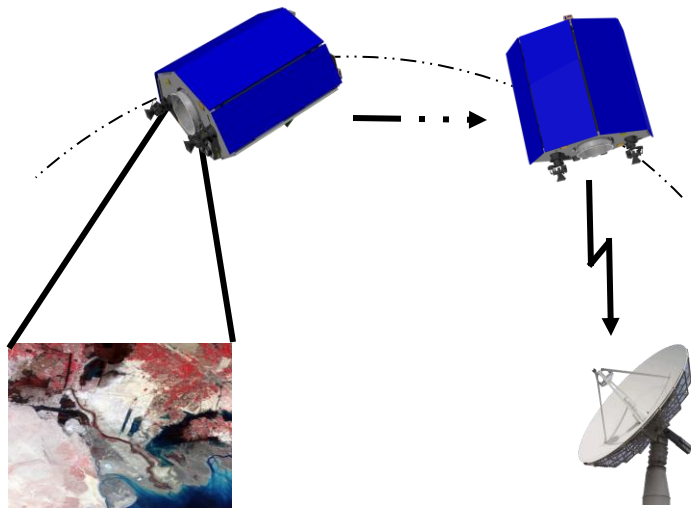


The apparent diameter, D , and angular size, ψ , can be related to the orbital height, h . The apparent diameter can usually be inferred from mission and payload requirements. The angular size is related to the pointing accuracy of the AOCS hardware (described later). These two can add an additional constraint to the orbit used.

Attitude Orbit Control Systems

Attitude Control Requirements: Rate of Attitude Change

There may be a requirement to change a satellites attitude, i.e., pointing at different targets for an Earth Observation satellite. This may require the spacecraft to change its pointing quickly, however, the system may have a maximum slew rate (or angular speed of rotation) which will act as a constraint on the system!



Remember that a **spinning mass** has **angular momentum, $\vec{H}$** [$\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$], which is a **function of its shape & mass distribution** (i.e. mass moment of Inertia, I [$\text{kg} \cdot \text{m}^2$]). The **rate of change in the angular momentum** is related to the **rate of change in the angular spin** (i.e. acceleration in its spin rate a.k.a. angular acceleration).

$$\dot{\vec{H}} = I\vec{\alpha}$$

Also note:

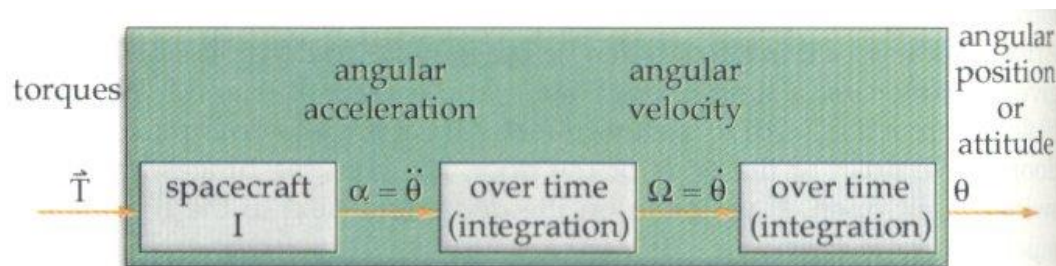
$$\vec{H} = I\vec{\Omega}$$

Note that torque is equivalent to the rate of change in angular momentum.

$$\vec{T} = \dot{\vec{H}} = I\vec{\alpha}$$

Also note $\vec{T} = \vec{R} \times \vec{F}$

Applying a torque to a free-floating object, will make the object spin faster by **providing angular acceleration**. An object having angular acceleration **gains angular velocity**, Ω , which will **change the objects attitude angle**, θ .



Attitude Orbit Control Systems

Attitude Control Requirements: Rate of Attitude Change

Note that the mass moment of Inertia, I [kg.m^2], can be described as a matrix. Rotations on a spacecraft occur around the principal axes. This is a co-ordinate system when the inertia products are equal to zero.

$$\mathbf{I} = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix}$$

$$\mathbf{I} = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix}$$

Products of inertia < 0

Moments of inertia > 0

Principal moments of inertia > 0

$$\mathbf{I}' = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

To compute the principal moments of inertia:

$$\mathbf{I} - \lambda \mathbf{1} = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Find the roots of the characteristic equation

$$(\mathbf{I} - \lambda \mathbf{1}) \cdot \bar{\boldsymbol{\Omega}} = \begin{bmatrix} I_{11} - \lambda & I_{12} & I_{13} \\ I_{21} & I_{22} - \lambda & I_{23} \\ I_{31} & I_{32} & I_{33} - \lambda \end{bmatrix} \begin{bmatrix} \Omega_x \\ \Omega_y \\ \Omega_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For example:

$$\mathbf{I} = \begin{bmatrix} 5 & 0 & -3 \\ 0 & 2 & 0 \\ -3 & 0 & 4 \end{bmatrix} \rightarrow (\mathbf{I} - \lambda \mathbf{1}) \cdot \bar{\boldsymbol{\Omega}} = \begin{bmatrix} 5 - \lambda & 0 & -3 \\ 0 & 2 - \lambda & 0 \\ -3 & 0 & 4 - \lambda \end{bmatrix} \begin{bmatrix} \Omega_x \\ \Omega_y \\ \Omega_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Expand out, use quadratic equations and then find the roots

$$\mathbf{I}' = \begin{bmatrix} 7.54 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1.46 \end{bmatrix}$$

Attitude Orbit Control Systems

Attitude Control Requirements: Rate of Attitude Change

Using the principal moments of inertia, we can calculate the changes in angular speed using Euler's equations for the motion of a rigid body under the influence of external torque.

For example:

Spacecraft has the following principal moments of inertia matrix

$$\mathbf{I}' = \begin{bmatrix} 40 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 30 \end{bmatrix}$$

Spacecraft has a current angular velocity matrix

$$\bar{\Omega} = \begin{bmatrix} 0.1 \\ 0.05 \\ 0.3 \end{bmatrix}$$

Then a torque to the spacecraft is applied

$$\vec{T}_C = \begin{bmatrix} 0 \\ 0 \\ -8 \end{bmatrix}$$

- 1) Extract angular acceleration (rate of change of angular velocity) for each axis.
- 2) Integrate over time to find the new angular velocity due to the applied torque for each axis.
- 3) Integrate over time to find the new angular position for each axis.

Hence you now have a new attitude.

Note your principal axis may not align with your reference co-ordinates and so attitude calculations can get even more complicated!

$$\dot{H}_1 = I_1 \alpha_1 = T_1 + (I_2 - I_3) \Omega_2 \Omega_3$$

$$\dot{H}_2 = I_2 \alpha_2 = T_2 + (I_3 - I_1) \Omega_3 \Omega_1$$

$$\dot{H}_3 = I_3 \alpha_3 = T_3 + (I_1 - I_2) \Omega_1 \Omega_2$$

Then: $\dot{H}_1 = I_1 \alpha_1 = 0 + (20 - 30) * 0.05 * 0.3 = -0.15 \text{ kg.m}^2.\text{s}^{-2}$

$$\dot{H}_2 = I_2 \alpha_2 = 0 + (30 - 40) * 0.3 * 0.1 = -0.3 \text{ kg.m}^2.\text{s}^{-2}$$

$$\dot{H}_3 = I_3 \alpha_3 = -8 + (40 - 20) * 0.1 * 0.05 = -7.9 \text{ kg.m}^2.\text{s}^{-2}$$

Then: $\alpha_1 = \frac{\dot{H}_1}{I_1} = \frac{-0.15}{40} = -3.75 \times 10^{-3} \text{ rad.s}^{-2}$

$$\alpha_2 = \frac{\dot{H}_2}{I_2} = \frac{-0.3}{20} = -0.015 \text{ rad.s}^{-2}$$

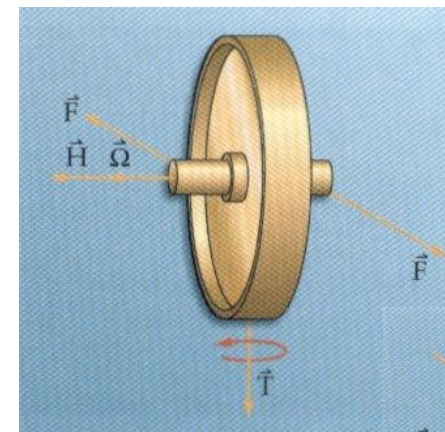
$$\alpha_3 = \frac{\dot{H}_3}{I_3} = \frac{-7.9}{30} = -0.263 \text{ rad.s}^{-2}$$

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Attitude Control Requirements: Gyroscopic Precession

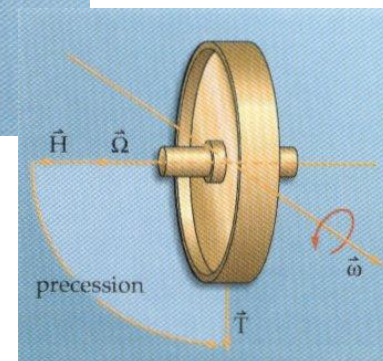
When torque is applied along the angular momentum vector then changes in angular acceleration and angular velocity occur (as previously discussed).

When torque is applied perpendicular to the angular momentum vector then the rotating object starts to rotate around an axis and precession occurs around the precession vector, ω , which is at right angles to the torque and angular momentum vectors. For a constant applied torque, the precession rate is also constant. The direction of precession depends on its mass moment of inertia.



Torque Applied
to a Spinning
Disk

Precession of a
Spinning Disk



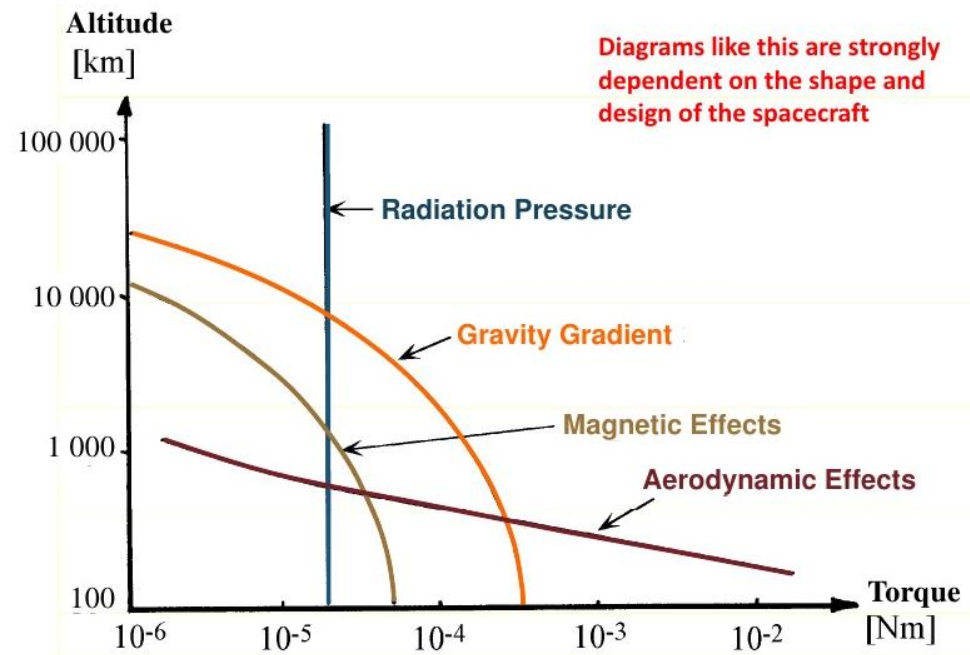
Attitude Orbit Control Systems

Disturbance Environment: What causes Torques?

External disturbance torques, come from four main sources. These are the Solar radiation pressure, Atmospheric drag, Magnetic torque due to a residual dipole and gravity gradient. These could be either **Secular** or **cyclical**. Cyclical disturbances average to Zero over a circular orbit and are analogous to conservative forces. Secular disturbances accumulate over the orbit and are analogous to non-conservative forces.

Internal disturbance torques can be generated on the spacecraft from Mechanisms (e.g. solar array drive, moving instruments etc.), Thermal snap, Misalignments and quantization in actuators, Propellant motion (i.e. sloshing), Other minor sources (e.g. astronauts (e.g. ISS), outgassing from polymers, differential IR emission).

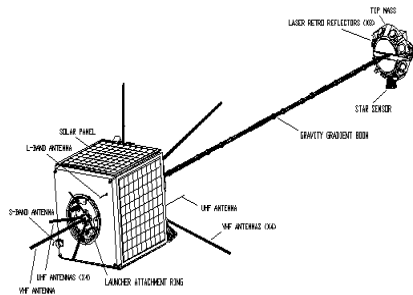
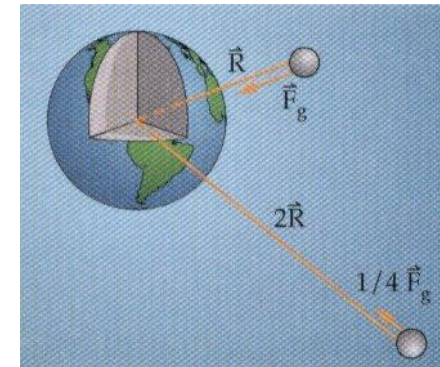
- Uncertainty in Spacecraft Center of Gravity (typically 1-3 cm)
- Thruster Misalignment (typically $0.1^\circ - 0.5^\circ$)
- Thruster Mismatch (typically $\sim 5\%$)
- Rotating Machinery
- Liquid Sloshing (e.g. propellant)
- Flexible structures
- Crew Movement



Attitude Orbit Control Systems

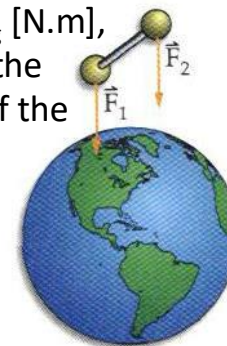
External Disturbance: Gravity Gradient

Consider two masses in the gravitational well of the Earth. One is double the distance of the other. The mass closest to the Earth will experience a Force four times greater than the mass furthest from the Earth.



Now consider two masses in the gravitational well of the Earth that are connected. The mass closest to the Earth will experience a larger force than the mass furthest away. This will create a Gravity-gradient torque, T_g [N.m], on the system, which will only settle down when the forces are aligned and pointing towards the foci of the gravity gradient well (a.k.a. local vertical).

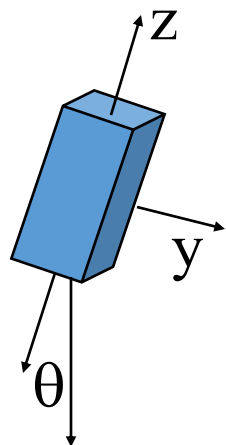
$$T_g = \left(\frac{3\mu}{2R^3} \right) |I_z - I_y| \sin(2\theta)$$



$F_1 > F_2$
this creates
a net torque
on the
spacecraft
→
which rotates
the spacecraft
back to vertical

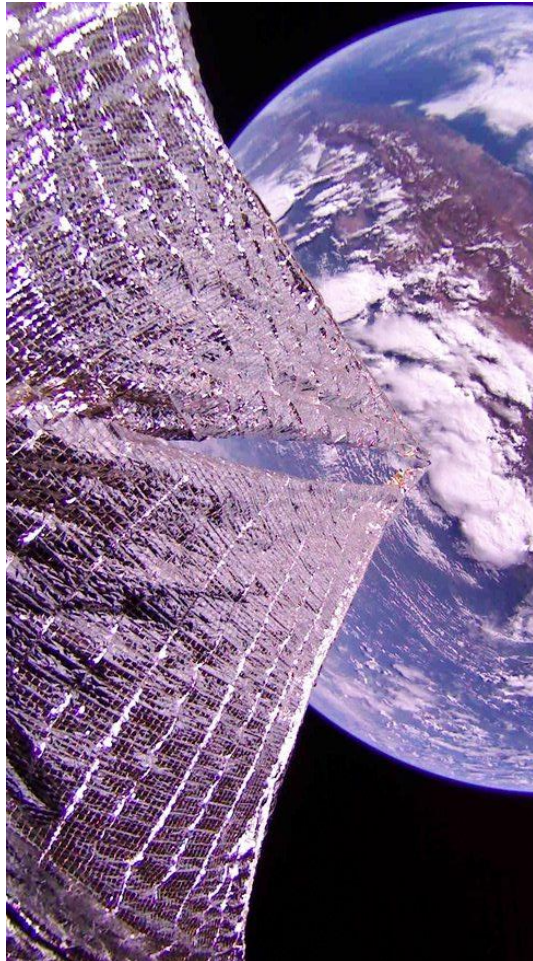


- The Gravity-gradient torque decreases with radius R^3 . So, the higher the orbit around a planet the less this effect has.
- The Gravity-gradient torque is only relevant on a spacecraft when one component of the moment of inertia is much greater than both the other components of moment of inertia i.e. $I_z \gg I_x = I_y$. If the moments of Inertia are homogeneous (i.e. $I_x = I_y = I_z$), this effect is zero.
- The greater the angle, θ , between the local vertical and the axis which has the largest moment of inertia associated with it (i.e. I_z) the greater the Gravity-gradient torque will be produced.

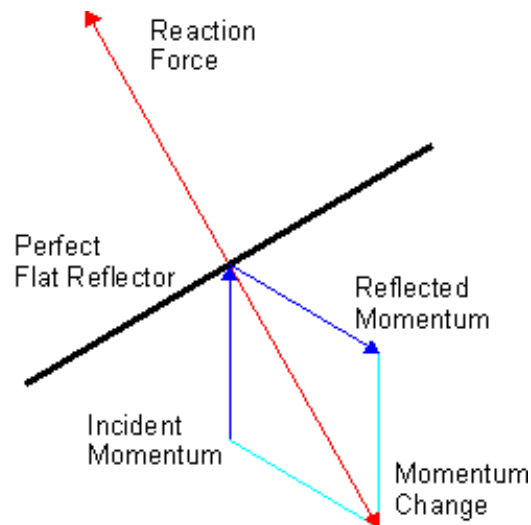


Attitude Orbit Control Systems

External Disturbance: Solar Radiation Pressure



Sunlight is made of tiny bundles of energy called photons. They are massless travelling at the speed of light, but unusually carry momentum. When photons strike a surface there is a small transfer of momentum (a.k.a. the Solar Radiation Pressure).



$$T_{srp} = F(c_{ps} - c_g)$$

$$F = \frac{F_s}{c} A_s (1 + \rho) \cos i$$

T_{srp} = solar radiation pressure disturbance torque

c_{ps} = center of solar radiation pressure

c_g = center of gravity

F_s = solar flux density $\left[\frac{\text{W}}{\text{m}^2} \right]$

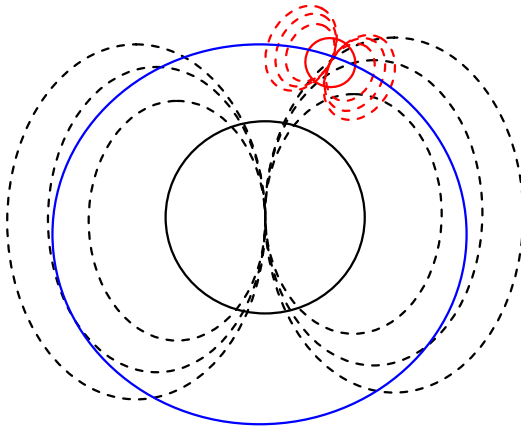
c = speed of light

A_s = area of illuminated surface

ρ = reflectance factor ($0 < \rho < 1$, typical value 0.6 for S/C)

i = sun incidence angle

External Disturbance: Planetary Magnetic Fields



A spacecraft can be perturbed by a planet's magnetic field in two significant forms, Spacecraft charging and magnetic dipole interactions.

$$\vec{T}_m = \vec{m} \times \vec{B}$$

Note value of m depends on S/C size and whether on-board compensation is used. Values can range from 0.1 to 20 Amp-m². $m = 1$ for typical small, uncompensated S/C

Spacecraft charging

- Spacecraft charging can occur when **charged particles from the solar wind**, interact with the spacecraft and 'deposit' their charge. If areas of the external structure are not properly grounded, then **charge can build up**. This may even lead to arcing (and cause damage in the process).
- The spacecraft will act like a point charge moving in a magnetic field and **create a Lorentz force**. This can perturb the orbit.
- In most cases this can be **eliminated by good grounding** of the Spacecraft.

Spacecraft magnetic dipole **Big concern for satellites in polar LEO**

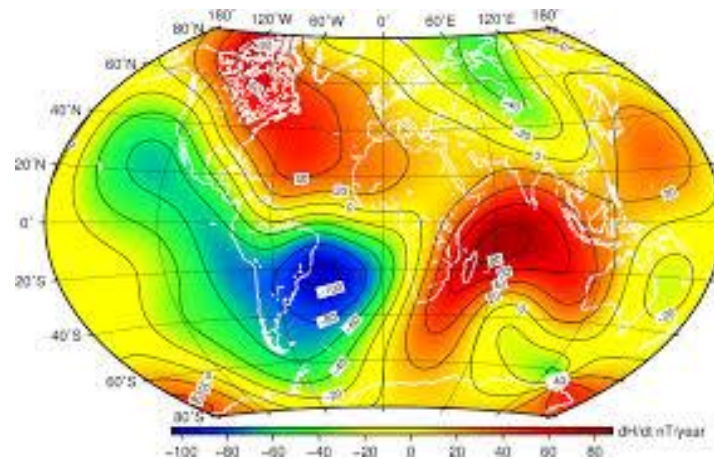
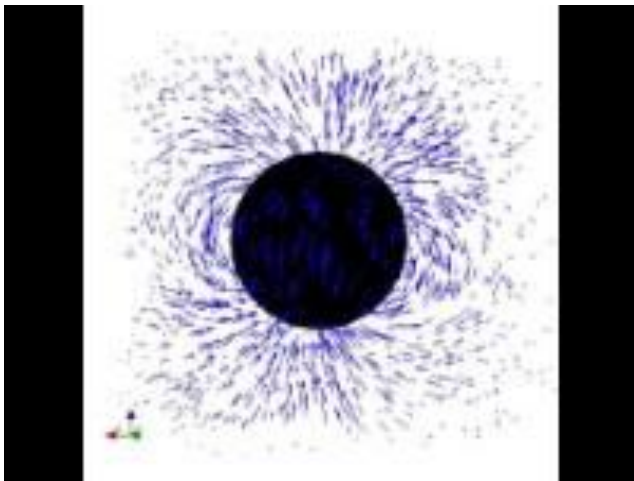
- Spacecraft have many **components** and made from many materials. These materials may **produce a small magnetic field** with a small magnetic dipole. **Closed current carrying circuit** loops can also create magnetic dipoles.
- When all these dipoles are summed together (remember they are vectors with differing orientations) then the spacecraft will have a **net effective magnetic dipole moment**, m [Amp.m²].
- The interaction of the spacecraft effective dipole moment and the Earth's magnetic field strength, B [Tesla], will create a **magnetic torque**, T_m . The Earth's magnetic field strength is complicated to model and **varies with distance** from the Earth's centre, R , and **latitude**.
- The magnetic dipole moment can be **reduced by careful selection of components** that do not use magnets and reducing closed carrying current loops (i.e. **use twisted pair wires**).

At the poles $B = \frac{M}{R^3}$ At the equator $B = \frac{2M}{R^3}$

Earth's magnetic moment, $M \approx 7.99 \times 10^{15}$ tesla.m³

Attitude Orbit Control Systems

External Disturbance: Planetary Magnetic Fields



$$\bar{T}_m = \bar{m} \times \bar{B}$$

Note value of m depends on S/C size and whether on-board compensation is used. Values can range from 0.1 to 20 Amp-m². $m = 1$ for typical small, uncompensated S/C

$$\begin{bmatrix} B_{north} \\ B_{east} \\ B_{down} \end{bmatrix} = \left(\frac{6378}{r_{km}} \right)^3 \begin{bmatrix} -C_\phi & S_\phi C_\lambda & S_\phi S_\lambda \\ 0 & S_\lambda & -C_\lambda \\ -2S_\phi & -2C_\phi C_\lambda & -2C_\phi S_\lambda \end{bmatrix} \begin{bmatrix} -29900 \\ -1900 \\ 5530 \end{bmatrix}$$

Where: C=cos , S=sin, ϕ =latitude, λ =longitude
Units: nTesla

Attitude Orbit Control Systems

External Disturbance: Atmospheric Drag



Various parts on a spacecraft could have different coefficients of drag, creating different forces acting upon the spacecraft's body. This causes there to be a difference in the location of the centre of atmospheric pressure and the centre of mass. This difference causes a drag torque.

$$T_a = F(c_{pa} - c_g)$$

$$\text{where: } F \cong \frac{1}{2} \rho C_D A v^2$$

T_a = aerodynamic disturbance torque

ρ = atmospheric density

C_D = coefficient of drag (typical S/C values are 2 - 2.5)

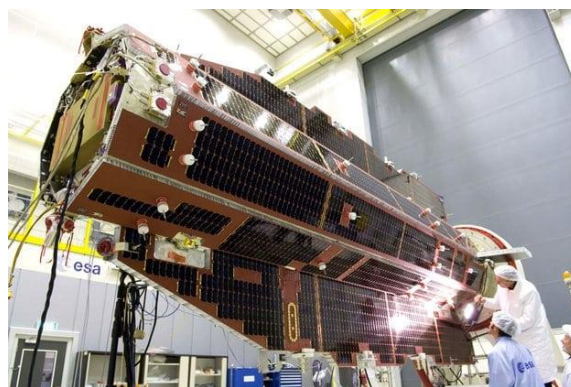
A = cross-sectional area

v = velocity

C_{pa} = center of atmospheric pressure

C_g = center of gravity

S/C designers can do little about it but will have to design a control system to cope



Attitude Orbit Control Systems

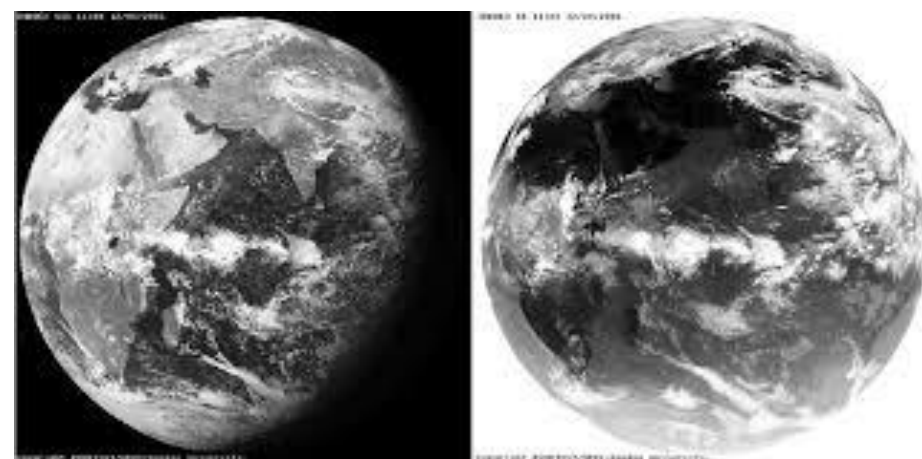
Attitude Determination

Attitude determination is the process of developing or **deriving an estimate of spacecraft attitude** from sensor measurement data. **Attitude knowledge is the level of certainty** associated with the attitude determination process. The magnitude of the uncertainty relies on sensor limitations, data quantization, sampling time, and numerical processing. **Real-time processing** on board the spacecraft **requires internal processing capabilities and computing resources**. **Post-processing** on the ground is **slower** but generally can provide a **definitive attitude** solution.

Sensor	Accuracies	Comments
IMU	Drift: 0.0003 – 1 deg/hr 0.001 deg/hr nominal	Requires updates
Star Sensor	1 arcsec – 1 arcmin (0.0003 – 0.001 deg)	2-axis for single star Multiple stars for map
Sun Sensor	0.005 – 3 deg 0.01 deg nominal	Eclipse
Earth Sensor GEO LEO	< 0.1 – 0.25 deg 0.1 – 1 deg	2-axis
Magnetometer	0.5 – 3 deg	< 6000 km Difficult for high <i>i</i>

Attitude Orbit Control Systems

Attitude Determination: Earth Sensors



An Earth Sensor (a.k.a. horizon sensor) uses IR to detect boundary between deep space & upper atmosphere (horizon). It uses IR as a visible image camera will get confused by the Earth's shadow.

In LEO, Earth fills a big portion of the sky

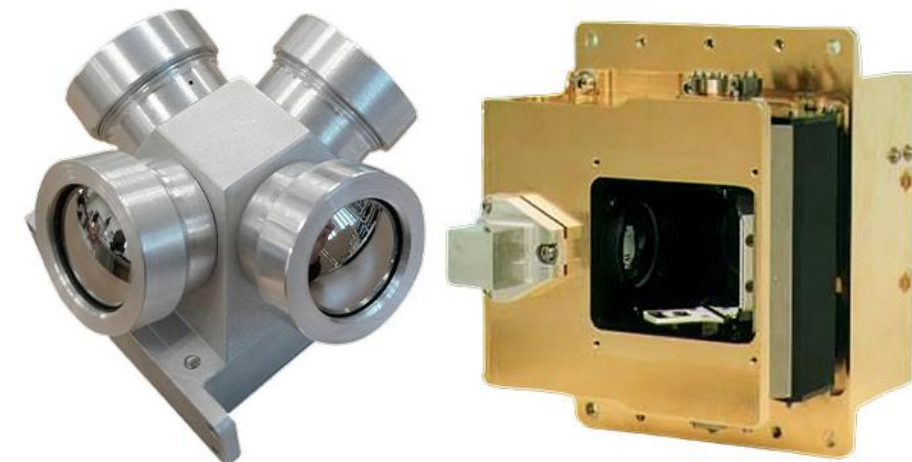
- Rough indication of “down” direction;
- Only discriminate where Earth is wrt rest of FOV

In GEO, angular radius of the Earth is $\sim 10^\circ$

- Sensor at least accurate to within that amount

For better accuracy, must focus only on one small portion

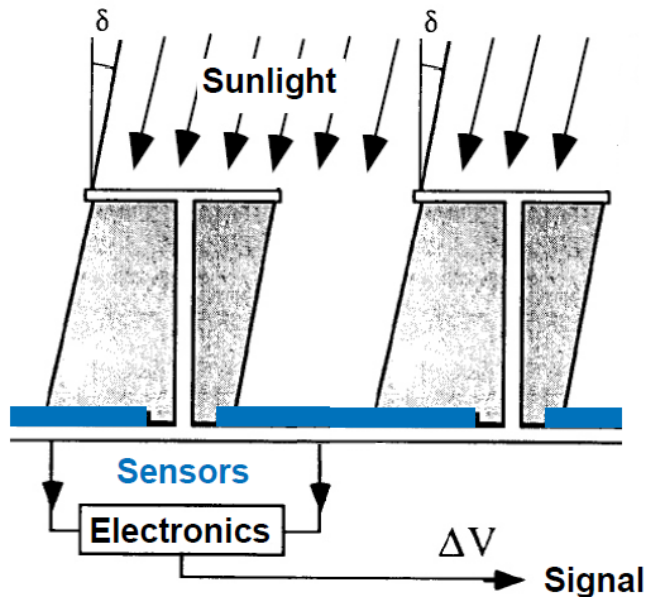
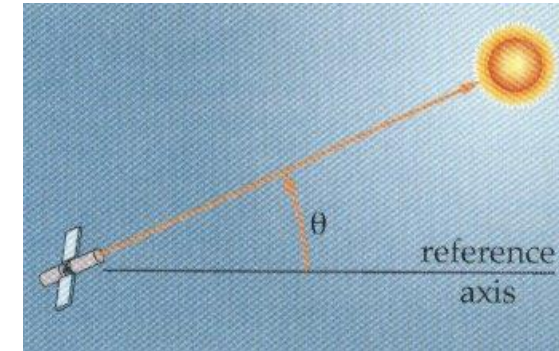
- Horizon detection: focus on atmospheric CO₂
- Up to 10× more accurate than Earth detector



Attitude Orbit Control Systems

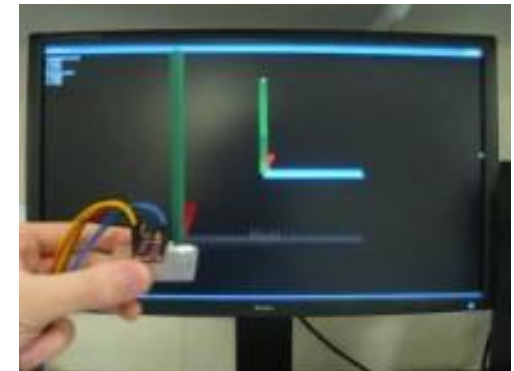
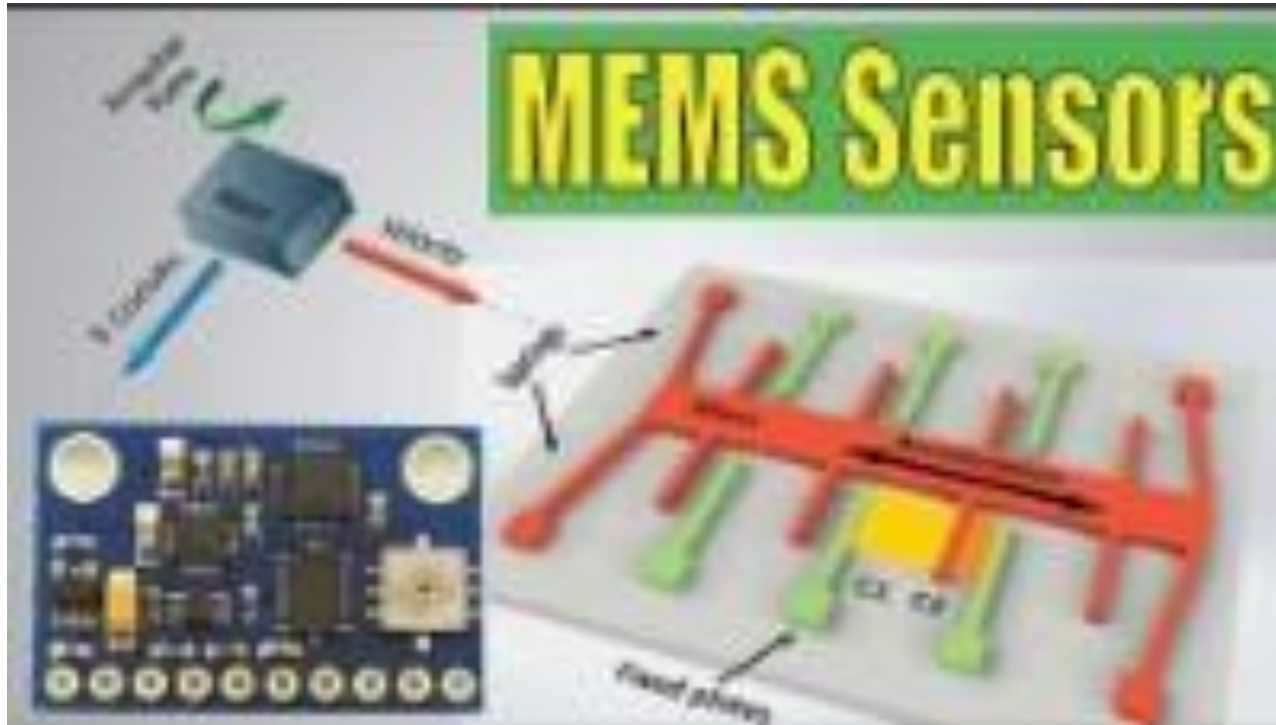
Attitude Determination: Sun Sensors

The Sun Sensor measures the position and direction of the Sun with respect to a body reference frame. It is the most widely used sensor.



Attitude Orbit Control Systems

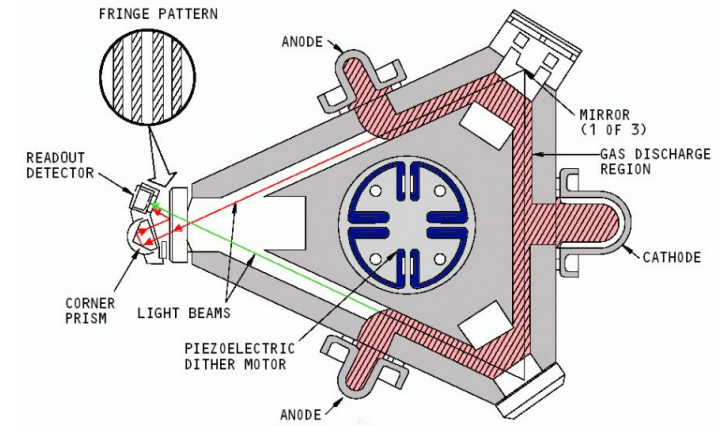
Attitude Determination: Inertial Measurement Unit



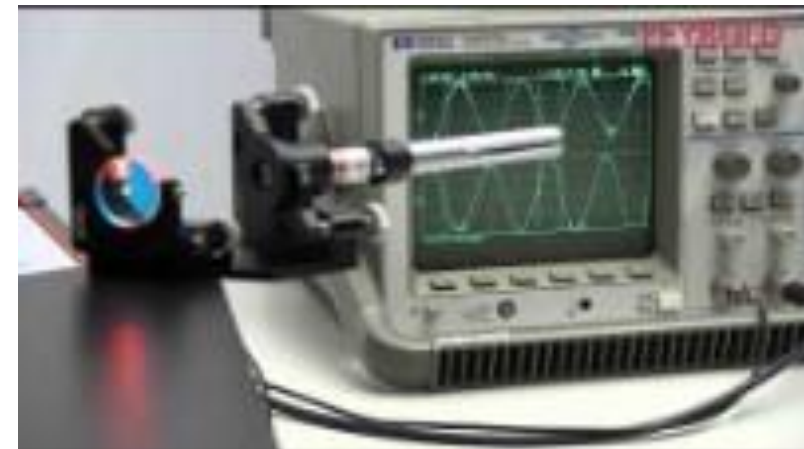
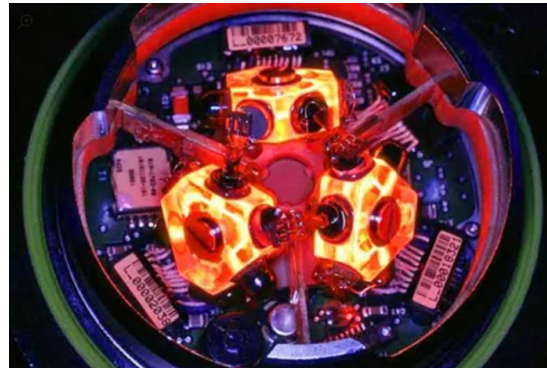
Modern IMUs contain MEMS accelerometers and gyroscopes, some even have inbuilt magnetometers. Measurements over time will drift so depending on accuracy the IMU needs to be updated regularly based on other data. Does not require knowledge of outside world to predict attitude.

Attitude Orbit Control Systems

Attitude Determination: Ring Laser Gyroscopes

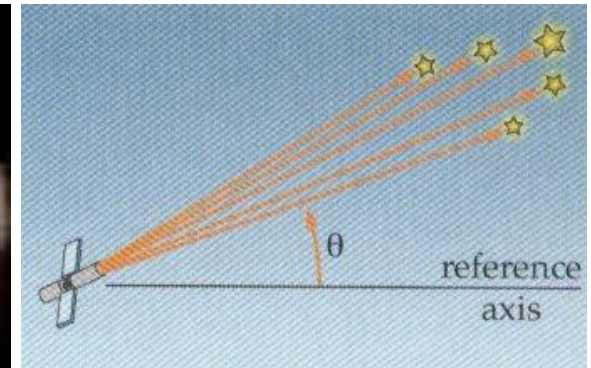


They typically “drift” about 0.0035 degrees per hour



Attitude Orbit Control Systems

Attitude Determination: Star Trackers



New research based on machine learning characterisation of star patterns.

Attitude Orbit Control Systems

Attitude Determination: GPS

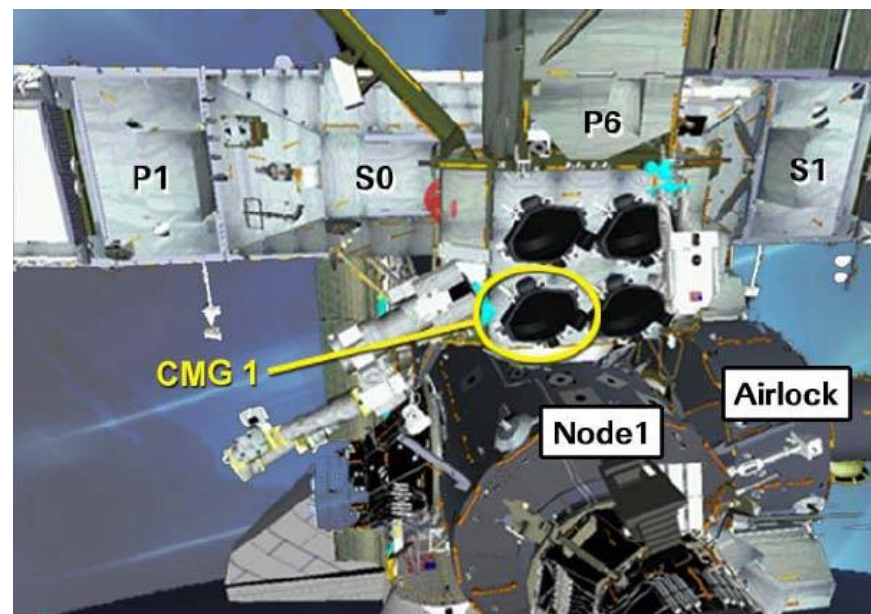


Attitude Orbit Control Systems

Attitude Control: Actuators

Passive – Attitude is maintained without control. They tend to be less accurate but less complex and more reliable. Examples include: Gravity gradient booms (rarely used), Dampers, Yo-yos, Spinning satellites (rarely used).

Active – Attitude is maintained using some form of control loop and active commanding by the system. They require constant feedback from attitude sensors. They are more complex but are more accurate in their pointing accuracy. Examples include; Thrusters, Reaction wheels, Gyros (Effective on large S/C (i.e. ISS, STS)), Magnetic Torque Rods.



Actuator	Accuracy	Lifetime limits	Comment
Gravity Gradient	$\pm 5^\circ$	None	2 Axis, Simple, Nadir pointing only
Spin Stabilized	$\pm 0.1^\circ$ to $\pm 1^\circ$	Propellant (if applies)	2 Axis, Rotation
Magnetic Torque Rods	$\pm 1^\circ$	None	High Current
Reaction Wheels	$\pm 0.001^\circ$ to $\pm 0.1^\circ$	Life of bearings	High Mass and Power, Momentum Dumping
Control Moment Gyro	$\pm 0.001^\circ$ to $\pm 0.1^\circ$	Life of bearings	High Mass and Power
Thrusters	$\pm 0.1^\circ$ to $\pm 1^\circ$	Propellant	Propellant limited, Large impulse

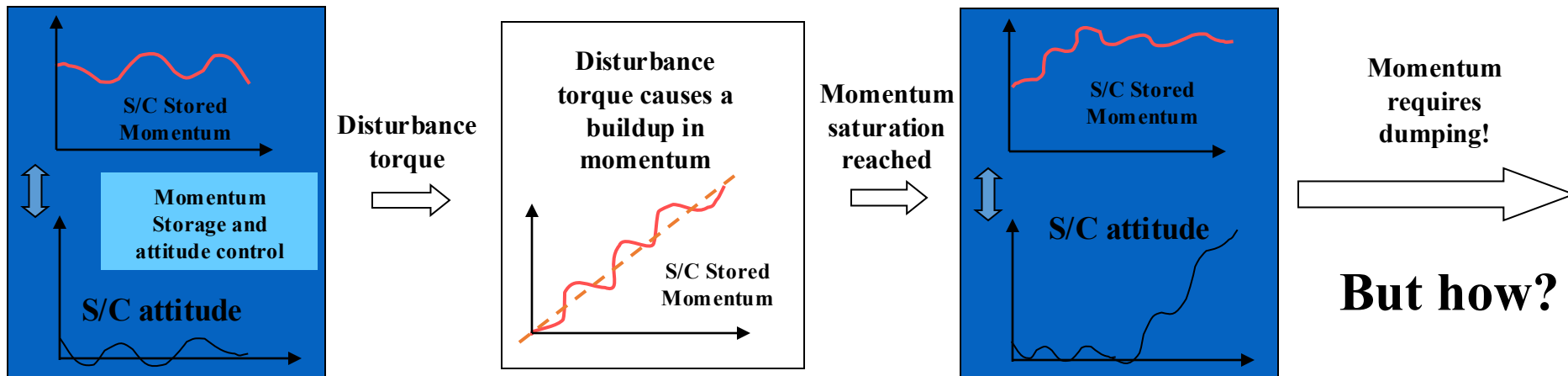
Attitude Orbit Control Systems

Attitude Control: Momentum Management



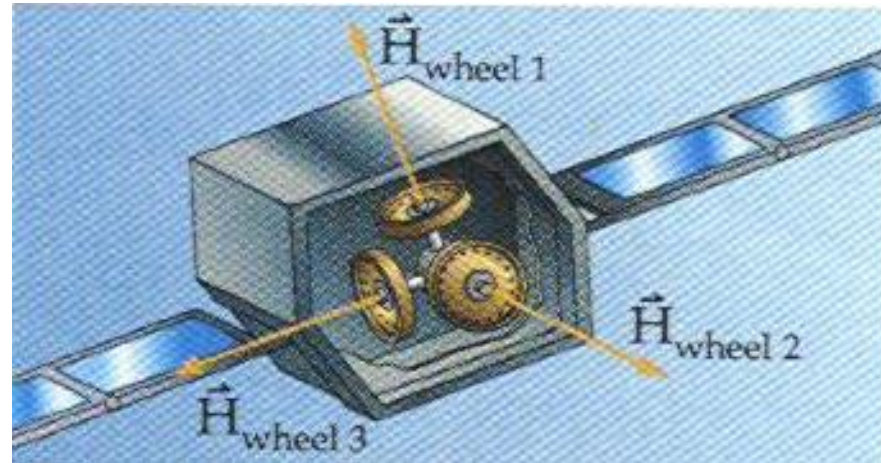
To precisely control the attitude of a spacecraft we effectively must manage the momentum acting on that spacecraft. Although this can be done by actuators directly (i.e. firing thrusters) most spacecraft tend to use reaction wheels.

Stored momentum can be thought of as a resource that the spacecraft can tap into, to precisely and effectively change the attitude by changing the rotation of the spinning reaction wheels and hence transferring momentum around the spacecraft. However, disturbance torques can add (and in some cases reduce) unwanted momentum which can cause issues...



Attitude Orbit Control Systems

Attitude Control: Reaction Wheels



Sometimes known as a Zero-bias system, reaction wheels try to maintain momentum of the spacecraft at (or close to) zero (i.e. keep the spacecraft pointing in one direction), by transferring and storing momentum in the spinning discs of the wheels.

Without external disturbances, the total angular momentum of the system is conserved, which is the angular momentum of the spacecraft body (i.e. the satellite body rotating) and the angular momentum stored in the rotating reaction wheels.

$$\vec{H}_{total} = \vec{H}_{S/C \text{ Body}} + \vec{H}_{wheel 1} + \vec{H}_{wheel 2} + \vec{H}_{wheel 3} + \dots$$

Also remember:

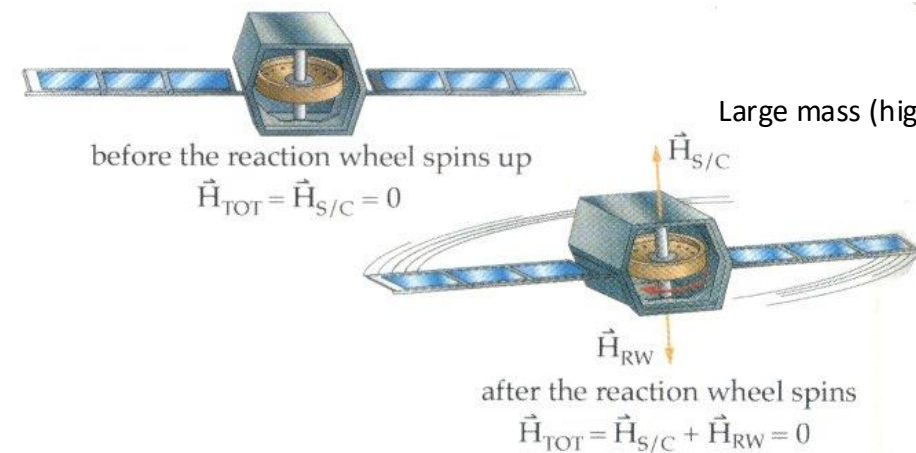
$$\vec{H} = I\vec{\Omega}$$

Large mass (high I) @ slow speed (low Ω) $\equiv$ small mass (low I) @ much higher rate (high Ω)

If a reaction wheel spins faster its angular momentum increases by ΔH_{RW} but H_{total} must stay constant. So the spacecraft body angular momentum must automatically decrease by an equal amount $-\Delta H_{S/C \text{ Body}}$, hence:

$$\Delta \vec{H}_{wheels} = -\Delta \vec{H}_{S/C \text{ Body}}$$

Attitude changes can be made by transferring angular momentum between the wheels and spacecraft body.



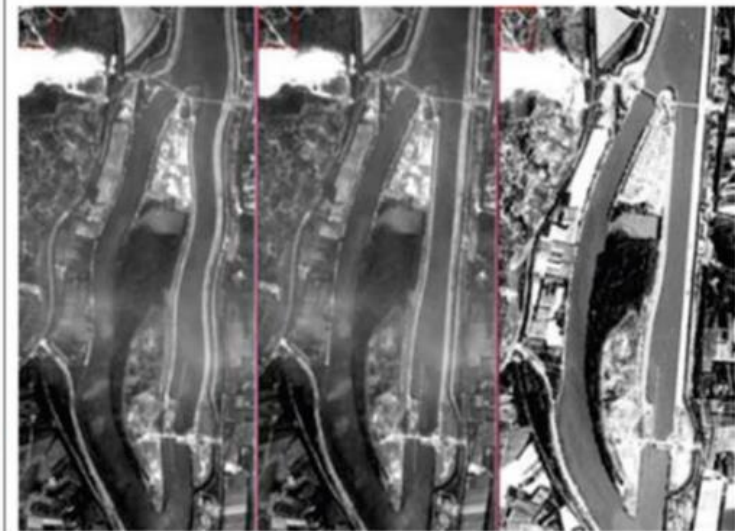
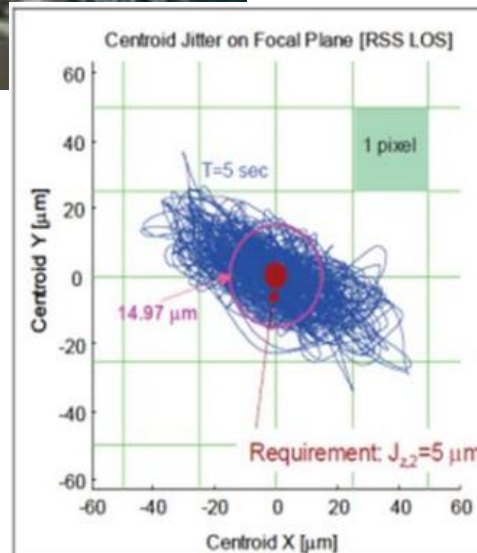
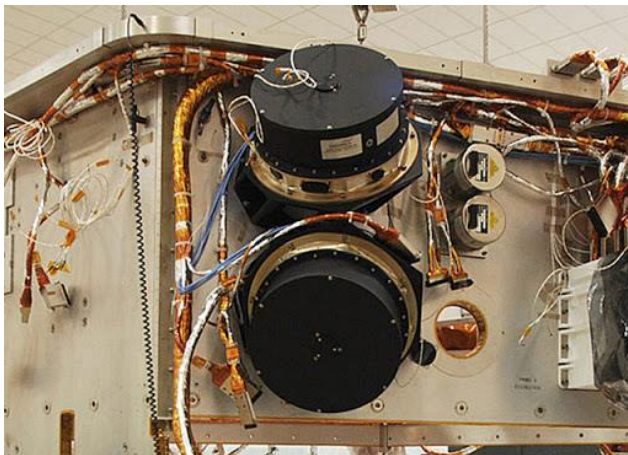
Attitude Orbit Control Systems

Attitude Control: Reaction Wheels

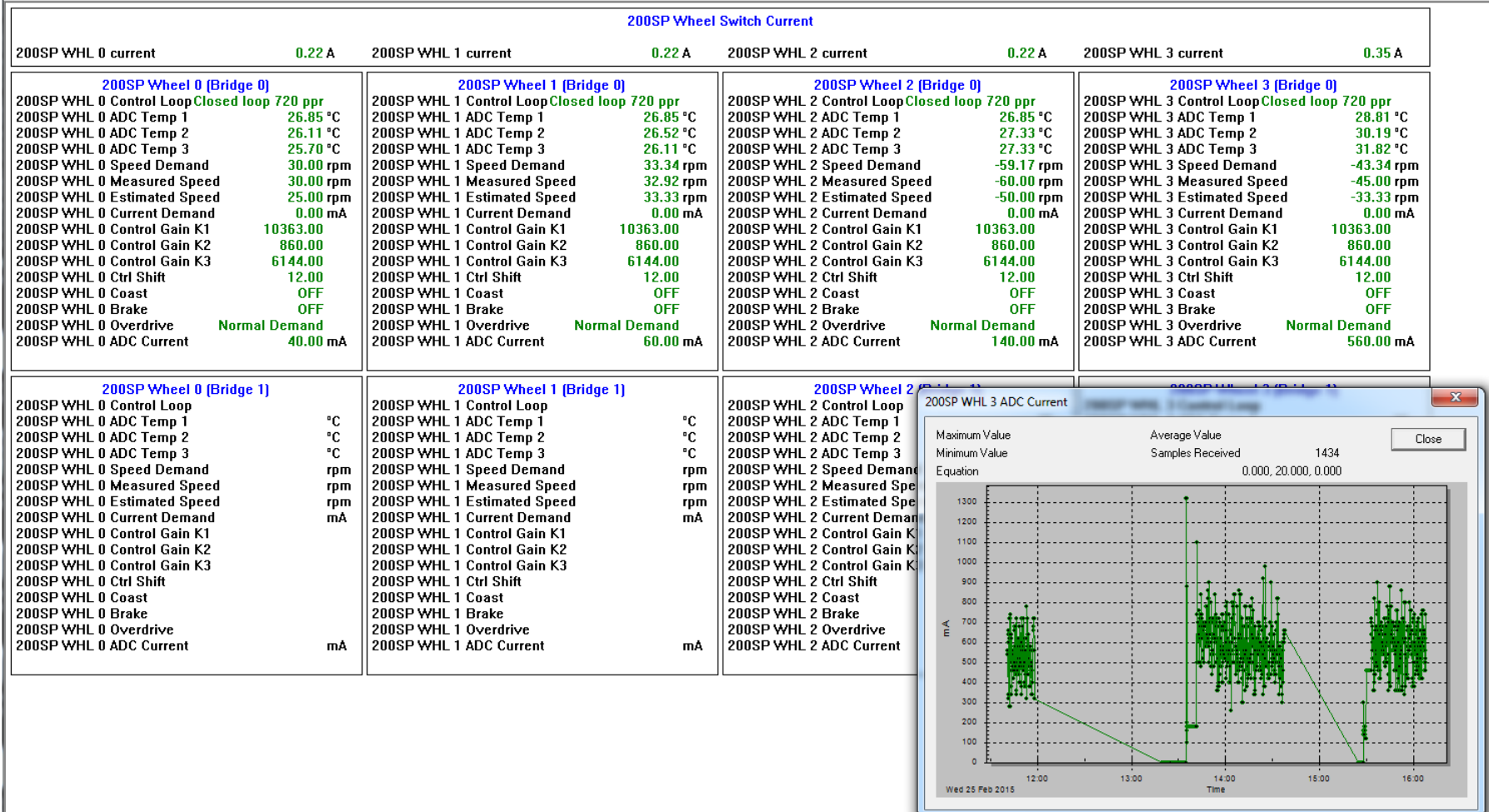


If space allows a 4 wheel system is typically used for redundancy purposes. Smaller satellites (CubeSATS tend to only have 3 or even just one).

Wear and tear of the fast spinning wheels can occur over time and so attention should take into account the types of wheel bearings and lubricants used as some are prone to out gassing and temperature effects. Lifetime tests are also very important.



Attitude Control: Reaction Wheels



Attitude Orbit Control Systems

Attitude Control: Reaction Wheels - Momentum Dumping

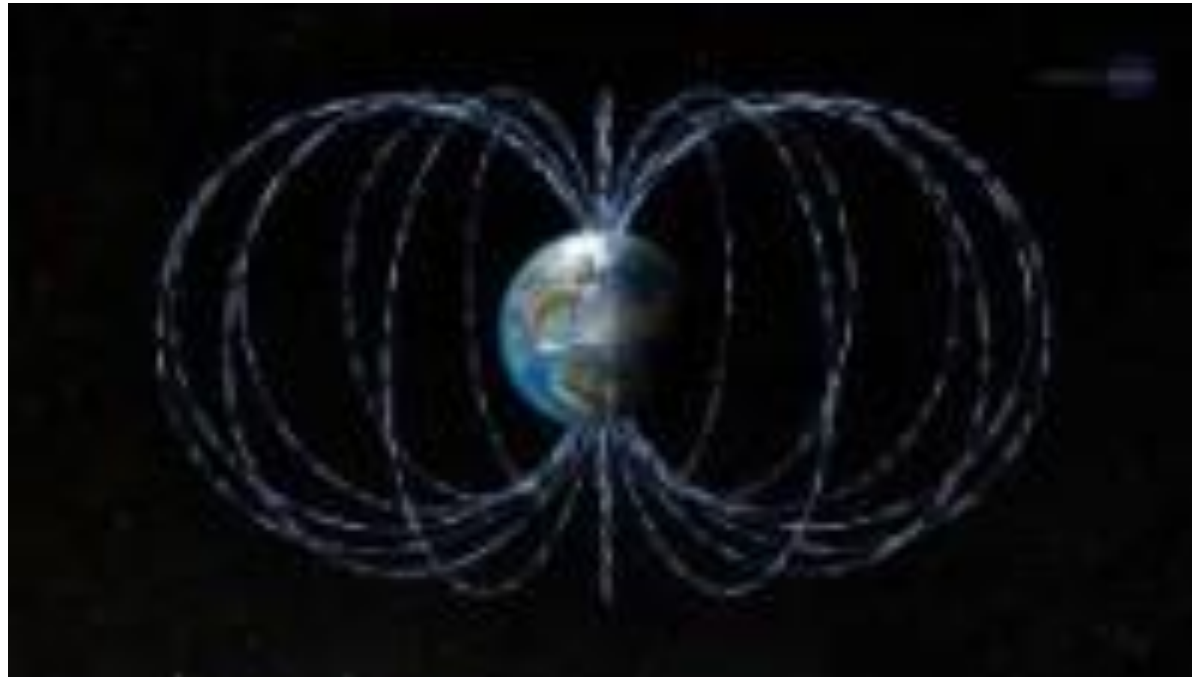
In a real system, there are constant disturbances on the spacecraft. This will impart unwanted torques on the system and in many scenarios increase the total momentum, H_{total} , stored. If the spacecraft is to maintain its attitude, the reaction wheels will need to spin faster to provide an opposing momentum to counteract the effect. However, reaction wheels have a physical limit to their maximum spin rotation. So when a reaction wheel system has spun as fast as it can and 'soaked' up as much momentum it can handle it is considered to be 'saturated'.

The system can be de-saturated through 'momentum dumping'. By applying a controlled torque, in the correct direction, using thrusters or magnetorquers, the angular momentum of the system can be decreased and hence by extension the spin rate of the wheels can be reduced.



Attitude Orbit Control Systems

Attitude Control: Magnetorquers

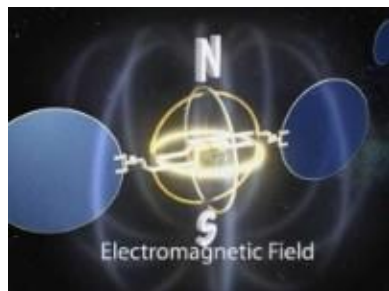


The magnetic dipole generated by the magnetorquer is expressed by the formula;

$$\overline{m} = nI\overline{A}$$

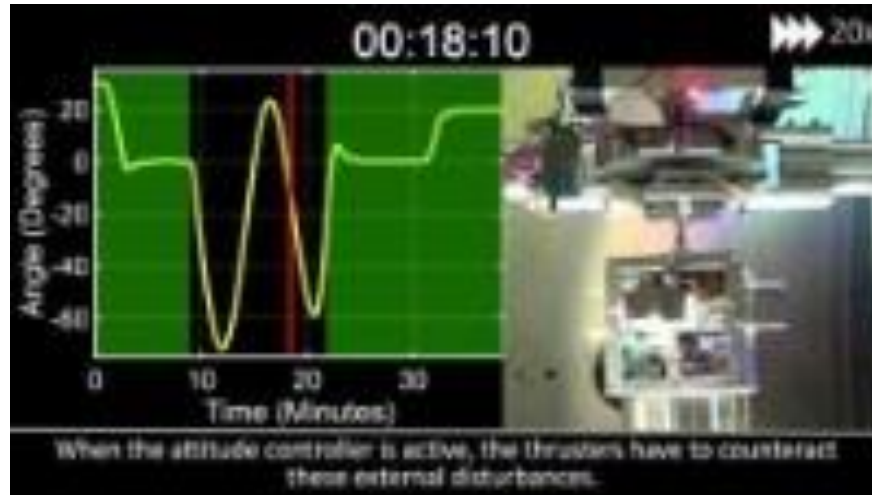
Where n is the number of turns of the wire, I is the current provided, and A is the vector area of the coil.

The magnetorquer rod is pulsed on and off and the dipole interacts with the Earth's magnetic field generating a torque. 3 magnetorquers in perpendicular angles to each other can be used to provide crude attitude control in LEO.



Attitude Orbit Control Systems

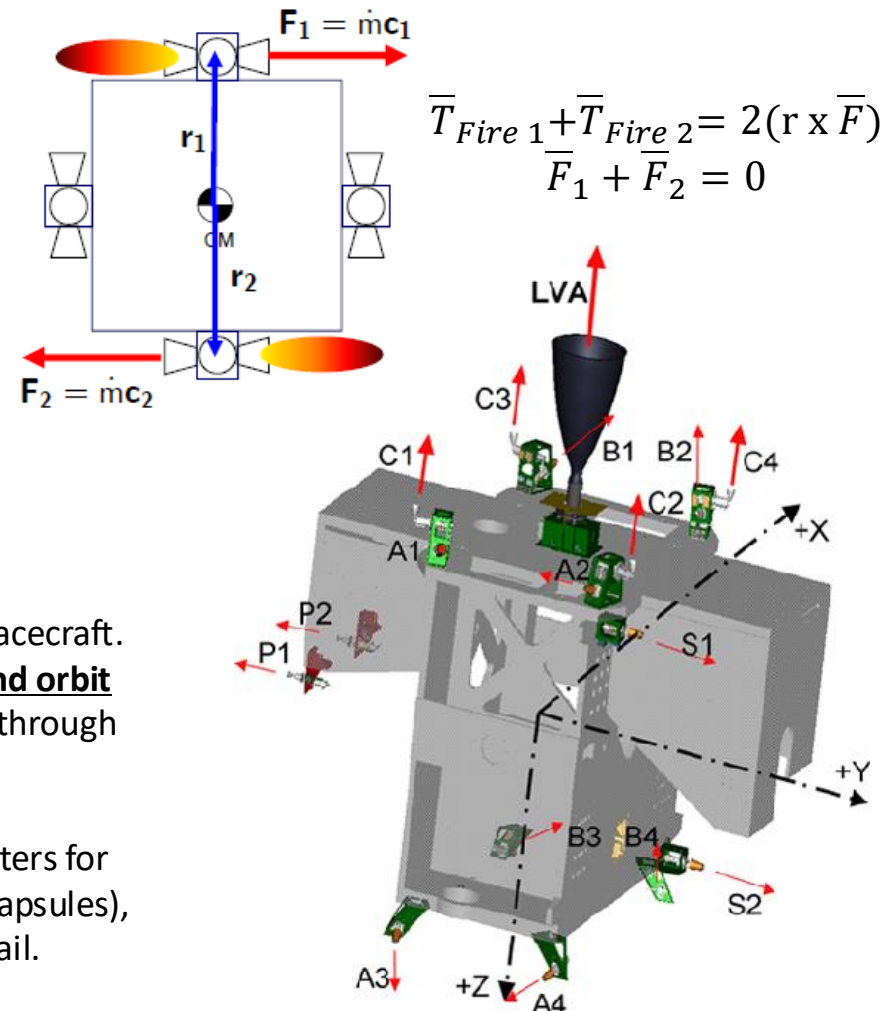
Attitude Control: Thrusters



Thrusters also known as reaction engines are used on most spacecraft. Depending on firing regime can be used for **attitude control and orbit control**. When used for attitude control, pairs of thrusters fire through the centre of gravity, causing a slew (i.e. rotation).

Limited by propellant, which is an expendable resource. Thrusters for attitude are used when there is plenty of supply (i.e. Space X capsules), for momentum dumping or as redundancy if reaction wheels fail.

Full explanation can be seen by watching this online lecture <https://www.youtube.com/watch?v=lsclmInrpKM>



Attitude Orbit Control Systems

Attitude Control: Yo-Yos



Yo-yo's are one time deployable masses on extendable tethers. The tethers are rapidly deployed and momentum is transferred from the spinning spacecraft body into the Yo-yo's. Yo-yos are operated in pairs to main attitude stability. Once the tethers are released it in effect reduces the total stored momentum, H_{total} , of the system.

Yo-yos do however cause space debris and in march 2009, a leftover yo-yo weight caused a scare when it came too close to the International Space Station.

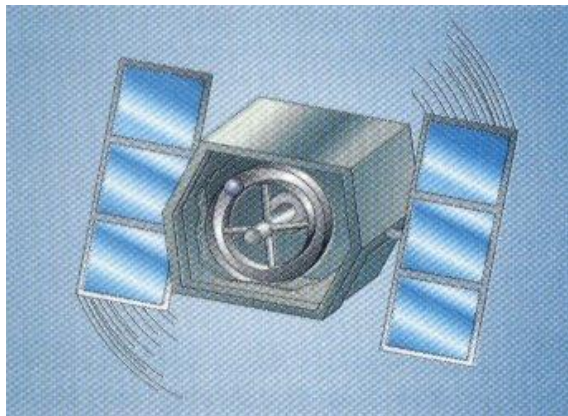
Attitude Orbit Control Systems

Attitude Control: Dampers



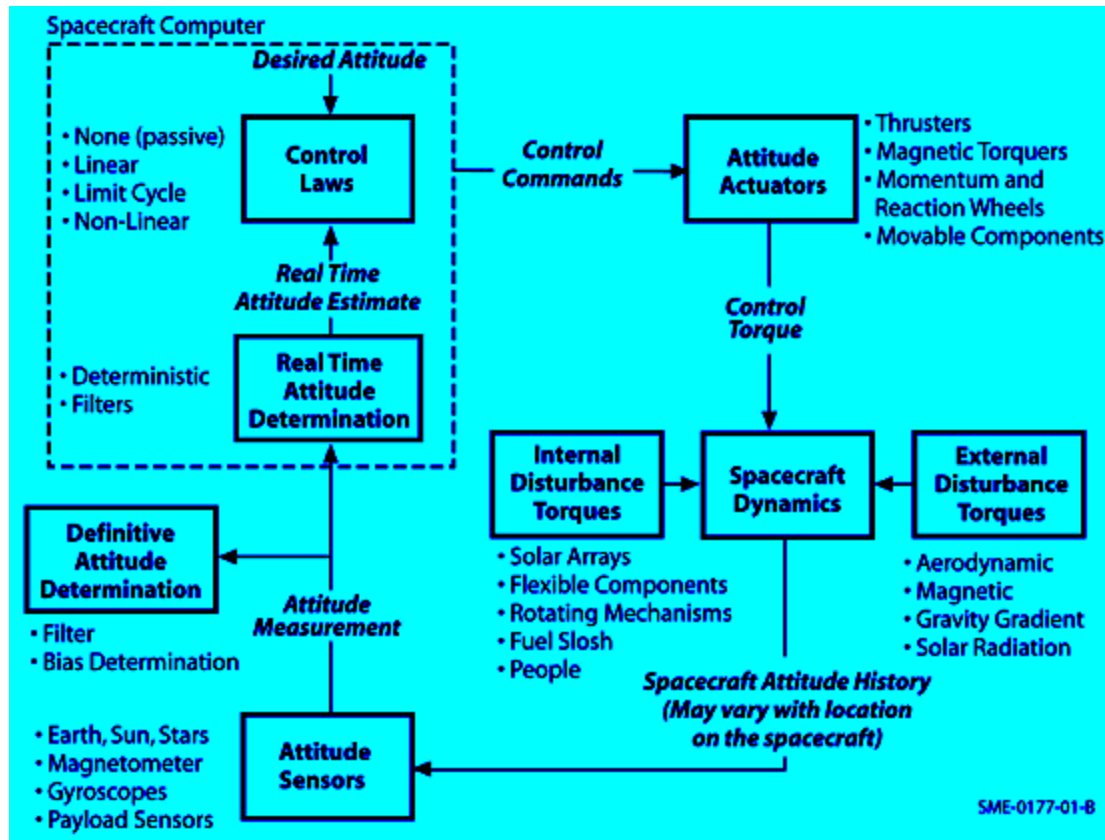
A damper changes the momentum by absorbing some of it and converting the energy, typically through friction, to other means. Most often the device is used in combination with other control systems.

Typical devices include; mass on springs, rubber structures, balls in circular tube filled viscous fluid etc. Friction between the ball and the fluid causes energy to be lost as heat and a reduction in overall momentum. A damper is usually designed to reduce disturbance around a single axis. In a spin stabilised spacecraft the dampers are used to remove unwanted wobbles in the spin axis. In EO satellites dampers are used to absorb micro vibrations. In launch vehicles dampers are used to keep them stable and upright on the launch pad.

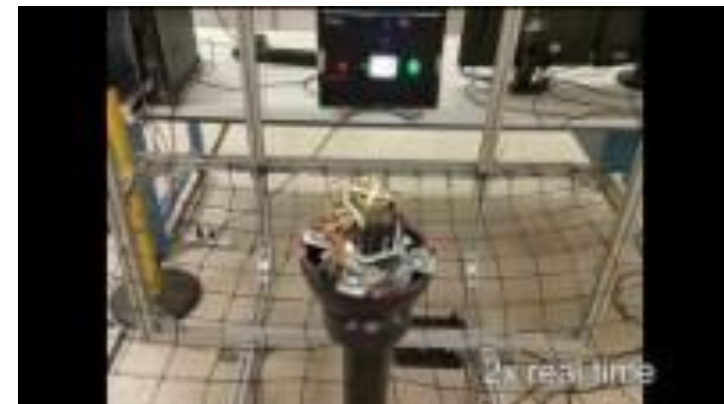


Attitude Orbit Control Systems

Attitude Determination Control (ADC): Subsystem Design



Mode	Description
Orbit insertion	Period prior to final orbit. Options: no control, spin stabilization of solid rocket motor, or full control using propulsion system
Acquisition	Initial determination of attitude and stabilization, following emergencies.
Normal on-station	Requirements for this phase drive design. Normal mission phases.
Slew	Re-orienting when required
Safe / contingency mode	Emergency situation if failure occurs, use less power and/or sacrifice normal operation to meet power and/or thermal constraints
Special	Special targets or times e.g. during eclipse



Attitude Orbit Control Systems

Attitude Determination Control (ADC): Specification

As a system engineer these are the minimum requirements you should try and establish

Attitude Determination Requirements

Accuracy	Orientation w.r.t. reference frame
Range	Range of angular motion over which accuracy is met

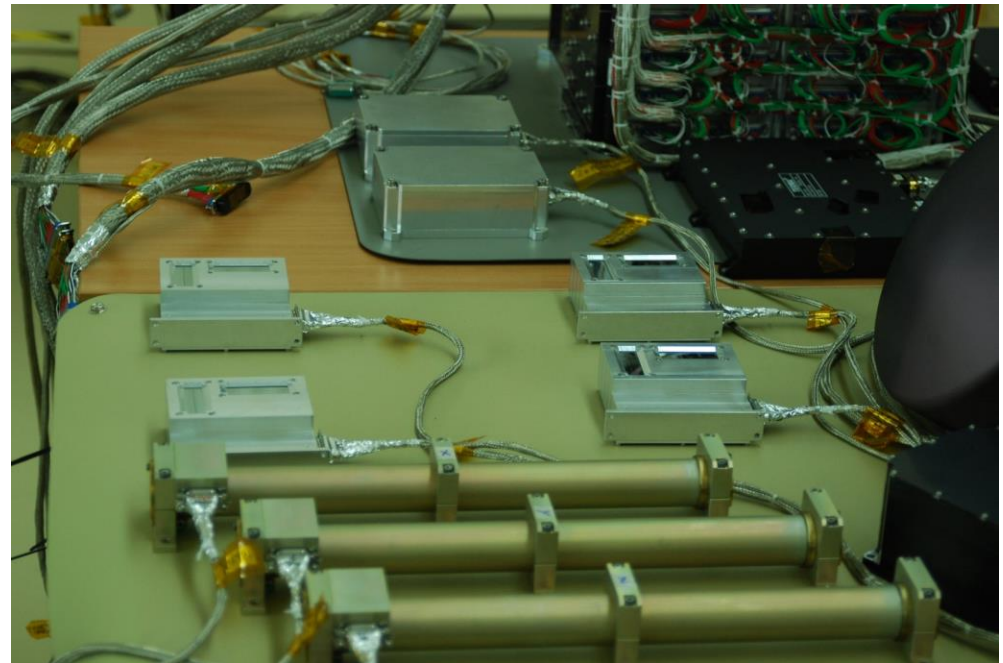
Attitude Control Requirements

Accuracy	Orientation w.r.t. reference frame
Range	Range of angular motion over which accuracy is met
Jitter	Limit on high frequency motion
Drift	Limit on slow, low frequency vehicle motion.
Settling Time	Time to recover from manoeuvres or upsets

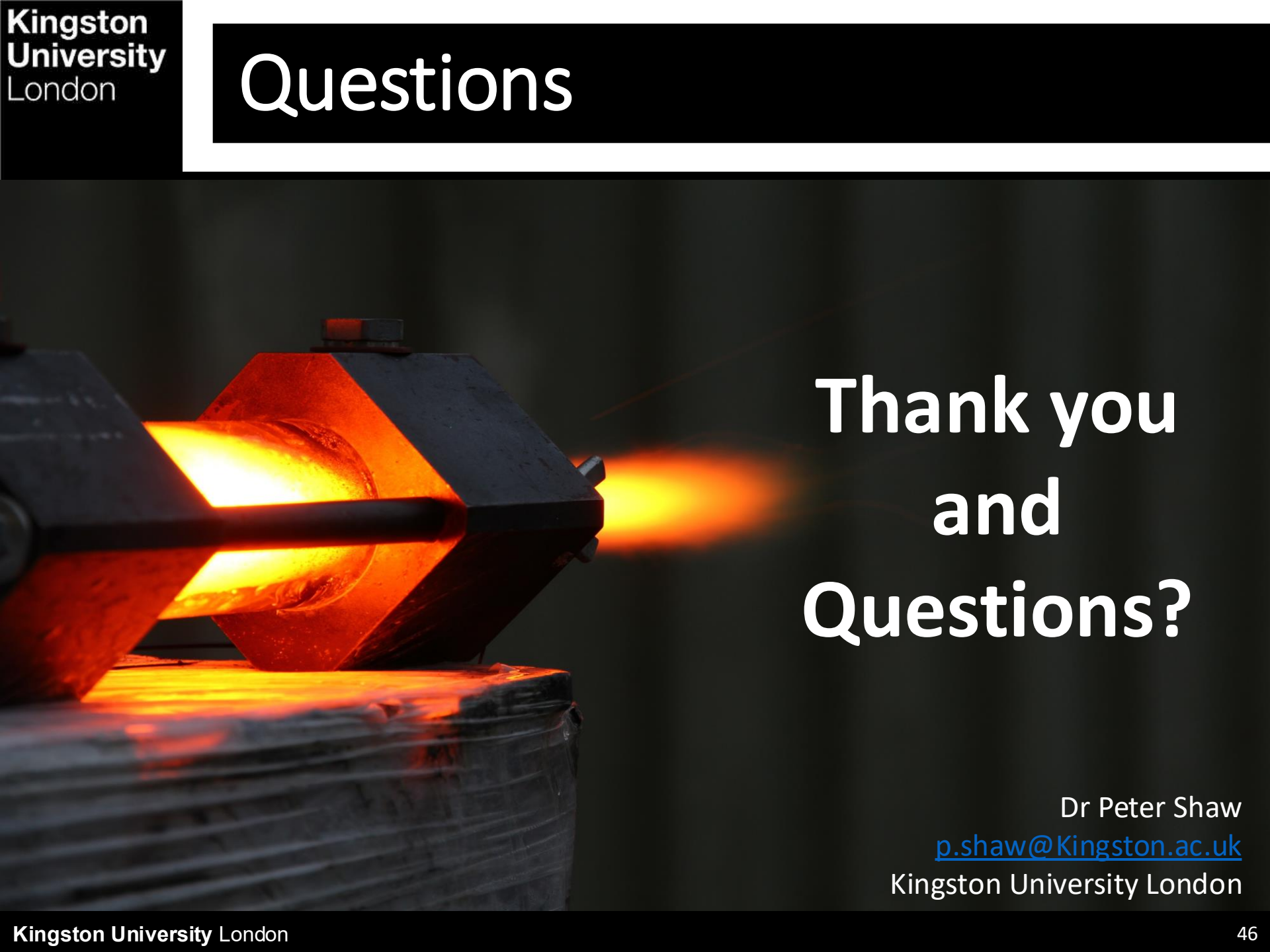
Step	Input	Output
1. Define modes and derive requirements	Mission requirements, profile, type of insertion	List of control modes, requirements & constraints
2. Select type of control for each mode	Payloads, thermal & power needs	Methods for stabilization & control
3. Quantify disturbances	Spacecraft geometry, orbit, solar/magnetic models, mission	Values for forces: gravity gradient, magnetic, aerodynamic, solar pressure, internal,
4. Select and size ADCS systems	Spacecraft geometry, pointing accuracy, direction, & slew rates, mission requirements, life time	Sensors: Actuators: Data processing: ...
5. Define ADC algorithms	As above	Algorithms, parameters,
6. Refine	As above	Final system specification

Attitude Orbit Control Systems

Attitude Determination Control (ADC): Specification



Questions

A close-up photograph of a glowing orange molten metal sphere being processed by industrial machinery. The sphere is held within a dark, angular metal frame, and a bright orange flame or spray of molten metal is visible to its right. The background is dark and out of focus.

**Thank you
and
Questions?**

Dr Peter Shaw

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